

LECTURE NOTE EEI 311. ELECTRICAL MEASUREMENT AND CONTROL

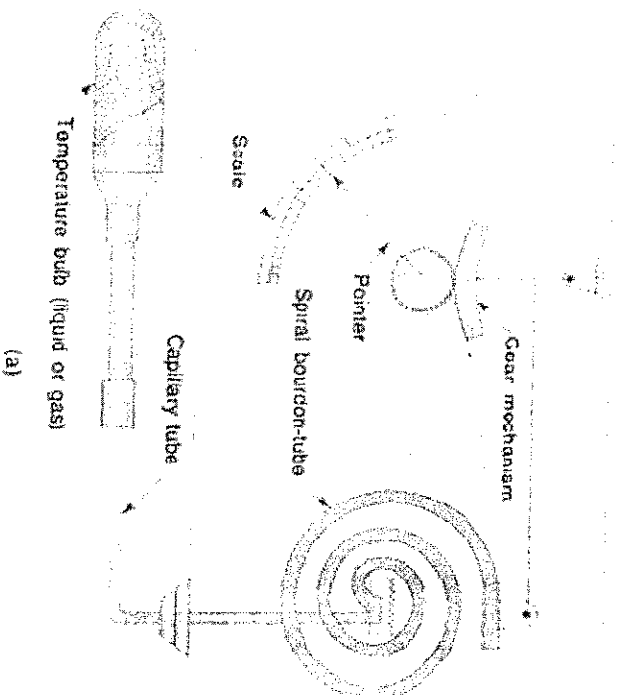
INTRODUCTION

Measurement can be said to be the process or act of finding the size, quantity or degree of something. It gives us a repeatable and dependable way of quantifying the world in which we live. It is essentially the act of comparing some unknown value with a value which is assumed to be known (otherwise known as a standard). Accurate measurement of quantities is very important in engineering. Instrumentation is when a set of instruments are used for a particular purpose (measurement or control). There are in engineering three ways in which instrumentation is used for making measurement.

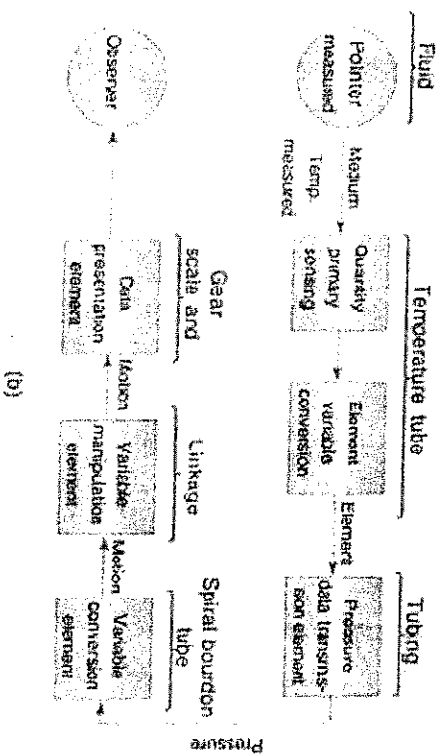
- 1) Obtaining data for some events or item: this could for instance be the making out of an item for machining and involve measurement of length and angles.
- 2) Inspecting an item to see if it matches the specification: in production engineering measurement are often concerned with determining whether an item being produced has the right dimension, shape, electrical resistance etc. that have been specified for the item.
- 3) Making measurement to ensure that a process is kept under control: many industrial processes are continuous. The purpose of measurement in this case is to ensure the proper control of the process.

Measurement systems (instruments): All measurement systems consist essentially of three main parts:

- 1) The sensing element: This frequently called the transducer is the first element. It is in some way in contact with what is being measured and produces a signal which is related to the quantity being measured. Sensing elements take information about the thing being measured and change it into some form which enables the rest of the measurement system to give a value to it. For example, if a spring balance is used to measure weight, the sensing action can be considered to be the change in length as a result of the weight. Thus, the spring balance takes information about the weight and changes, into a change in length of a spring.
- 2) The signal converter: The output of the sensing element then passes through a second element before reaching the display. This second element can take many forms. In general, it can be considered the signal converter in that the signal from the sensing element is converted into a form which is suitable for the display or control element. An example of this might be an amplifier which takes a small signal from the sensing element and makes it big enough to activate a display.
- 3) The display: The third part of the measuring system could be the display or a control system. The display element is where the output from the measuring system is displayed. This may, for example, be a pointer moving across a scale. The display element takes the signal from the signal converter and presents it in a form which enables an observer to recognize it. The control system is where the output is used to control a process.



(a)



(b)

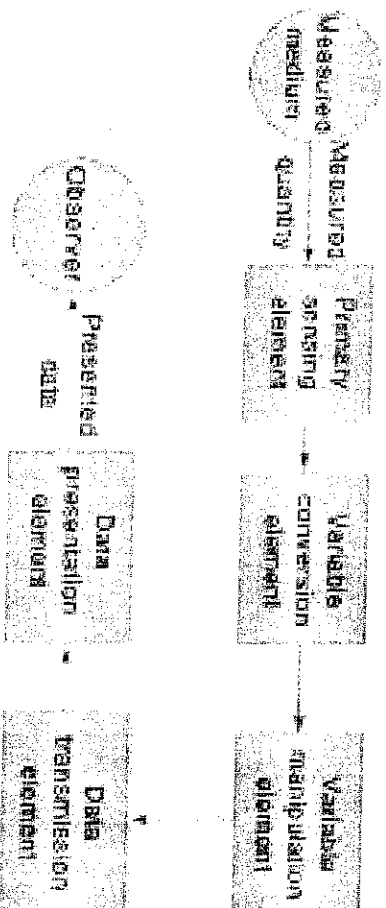


Fig. 1: fundamental elements of an instrument system

Performance characteristics

The performance characteristics of an instrument are very necessary for choosing the most suitable instrument for a specific measuring task. This generally has two sub areas:

static characteristics

dynamic characteristics

dynamic characteristics

Static characteristics: the static characteristics are generally considered for instrument used in measuring unvarying process condition. These characteristics are usually obtained by one form or another of a process called **calibration**.

Calibration may be defined, in general, as the process for determination, by measurement or comparison with a standard, of the correct value each scale reading on a meter or measuring instrument; or determination of the settings of a control device respond to particular values of volt current, frequency, pressure, flow or some other output.

In performing a calibration of an instrument, the following steps are necessary

Examine the construction of the instrument and identify and list possible inputs.

Decide, as best as one can, which of the inputs will be significant application for which the instrument is to be calibrated.

Procure apparatus that will allow all significant inputs to vary ranges considered necessary.

By holding some inputs constant, varying others, and recording the develop the desired static input-output relations.

the following are some of the terms commonly used to describe the performance of measuring systems:

Accuracy: the accuracy of an instrument is the extent to which the reading it gives might be wrong or the ability of a device or a system to respond to the true value of a measured variable under reference conditions. It is usually quoted as a percentage of the full-scale deflection (f.s.d) of the instrument. Thus, for example, an ammeter might have an f.s.d. of 5A and accuracy quoted as $\pm 5\%$. This means that 0.25 A is either added or subtracted from any reading taking on this ammeter. Accuracy can either be static for slow changing quantities or dynamic for quantities that changes quickly.

Precision: this is the degree of exactness for which an instrument is designed to perform. It consists of two characteristics, conformity and the number of significant figures to which the measurement may be made. The more the significant the figures, the greater the precision of the instrument.

Repeatability: the repeatability of an instrument is its ability to display the same reading for repeated applications of the same value of the quantity being measured.

Reliability: the reliability of an instrument is the probability that it will operate to an agreed level of performance under the conditions specified for its use.

Reproducibility: the reproducibility or stability of an instrument is its ability to display the same reading when it is used to measure a constant quantity over a period of time or when that quantity is measured on a number of occasions.

Sensitivity: this can be defined as the ratio of the change in output to the change in input which causes it, at steady state conditions. the sensitivity of an instrument is given by

$$\text{sensitivity} = \frac{\text{change in instrument scale reading}}{\text{change in the quantity being measured}}$$

Thus, for example, a volt meter might have a sensitivity of 1 scale division per 0.05V. This means that if the voltage being measured changes by 0.05V, then the reading of the instrument will change by one scale division.

Resolution: the resolution or discrimination of an instrument is the smallest change in the quantity being measured that will produce an observable change in the reading of the instrument.

Range: The range of an instrument is the limits between which readings can be made

Dead space: The dead space of an instrument is the range of values of the quantity being measured for which it gives no reading.

Threshold: If the quantity being measured is increased from zero a certain minimum level might have to be reached before the instrument responds and gives a detectable reading. This is called the threshold.

Hysteresis: Instruments can give different reading for the same value of measured quantity depending on whether that value has been reached by a continuously increasing change or continuously decreasing change. This effect is called hysteresis and it occurs as a result of such things as bearing friction and slack motion in gears in instrument.

Dynamic characteristics: instruments rarely respond to changes in measured variables, rather they exhibit a characteristic of slowness or sluggishness due to things as mass, thermal capacitance, fluid capacitance or electric capacitance. In addition to this, pure delay in time is often encountered when the instrument waits for some reactions to take place. the dynamic behavior of an instrument is determined by subjecting its primary element to some unknown and predetermined variations in measured quantity. the three most common variations are:

- Step change in which the primary element is subjected to an instantaneous and finite change in measured variable
- Linear change in which the primary element is following a measured variable changing linearly with time
- Sinusoidal change in which the primary element follows a measured variable the magnitude of which changes in accordance with the sinusoidal function of a constant amplitude.

The dynamic characteristics of an instrument as follows:

Speed of response: it is the Rapidity with which an instrumental response to changes in the measured quantity

Fidelity: it is the degree to which an instrument indicates the changes in the measured variable without dynamic error

Lag: it is retardation or delay in the response of an instrument changes in the measured quantity.

Dynamic error: it is the difference between the true value of a quantity changing with time and the value indicated by the instrument if not static error is assumed.

Error = measured value - true value

Quantity	Unit	Unit Symbol
Fundamental (Basic) Units		
Length	Meter	m
Mass	Kilogram	kg
Time	Second	s
Electric Current	Ampere	A
Thermodynamic Temperature	Kelvin	K
Luminous Intensity	Candela	cd
Supplementary Units		
Plane Angle	Radian	rad
Solid Angle	Steradian	sr
Quantity of Substance	Mole	mol
Derived Units		
Area	Square metre	m ²
Volume	Cubic metre	m ³
Velocity	Metre per second	m/s
Angular Velocity	Radian per second	rad/s
Frequency	Hertz	Hz
Density	Kilogram per cubic metre	kg/m ³
Acceleration	Metre per second squared	m/s ²
Angular Acceleration	Radian per second squared	rad/s ²
Force	Newton	N (kg m/s ²)
Pressure, Stress	Newton per square metre	N/m ²
Power	Watt	W (J/s)
Work, Energy	Joule	J (Nm)
Potential difference, Electromotive Force	Volt	V (W/A)
Electric Capacitance	Farad	F (A s/V)
Electric Resistance	Ohm	Ω (V/A)
Magnetic Flux	Weber	Wb (V s)
Inductance	Henry	H (V s/A)
Electric Field strength	Volt per metre	V/m
Magnetic Field strength	Ampere per metre	A/m
Luminous Flux	Lumen	lm (cd sr)
Luminance, Brightness	Candela per square metre	cd/m ²
Illumination	Lux	lx (lm/m ²)

Fundamental and derived quantities

Multiple	SI Prefix	Symbol
10 ¹²	tera	T
10 ⁹	giga	G
10 ⁶	mega	M
10 ³	kilo	k
10 ²	hecto	h
10	deka	da
10 ⁻¹	deci	d
10 ⁻²	centi	c
10 ⁻³	milli	m
10 ⁻⁶	micro	μ
10 ⁻⁹	nano	n
10 ⁻¹²	pico	p
10 ⁻¹⁵	femto	f
10 ⁻¹⁸	atto	a

Exercises:

- 1) A hydrometer is specified as having a range of 600-650 kg/m³ and an accuracy of ± 0.5 kg/m³. Explain the significance of this data.
- 2) An ammeter is quoted to have a range of 0-3 A and an accuracy of $\pm 2\%$ f.s.d. what is the accuracy which can be quoted for a current reading of 2 A?
- 3) A thermocouple is specified as having a sensitivity of 0.03 millivolt/°C. What does this mean?
- 4) A force measuring system has a range of 0-200N with a resolution of 0.1% f.s.d. what is the smallest change in force that can be measured?

TEMPERATURE MEASURING SYSTEMS

Temperature: the temperature of a substance is a measure of the hotness, or the coldness of that substance. It is the thermal state of the body which determines whether it will give heat to, or receive heat from other bodies. The term temperature and heat are closely related. Temperature may be defined as “degree of heat” but heat is usually taken to mean “quantity of heat”. Temperature and heat are related quantitatively since heat flows, of its own accord, from a body at higher temperature to a body at a lower temperature. It is therefore important to remember that in temperature measurement, two bodies in intimate contact are at the same temperature only if there is no heat flow between them.

Temperature scales: Temperature scales are based upon some recognized fixed points. At least two fixed points are required which are constant in temperature and can be easily reproduced such as:

the lower fixed point, or ice-point

the upper fixed point, or steam-point

The lower fixed point, or ice point, is the temperature of ice, prepared from distilled water, when melting under a pressure of 760 mm hg. The upper fixed point is the temperature of steam from pure distilled water boiling under a pressure of 760 mm hg. The boiling point of water varies greatly with applied pressure. Thus, it is important to note the pressure at which the water is boiling.

The temperature difference between the ice point and the steam point is known as the “fundamental interval”. In order to graduate a thermometer between these fixed points, the temperature between these points is divided into a number of equal parts.

Fahrenheit and centigrade (Celsius) temperature scales: the Fahrenheit scale, abbreviated °F, was introduced in 1709 by a German philosopher Fahrenheit. On this scale, the melting point of ice is designated as 32°F and the boiling point at 212°F. The centigrade scale was introduced in 1742 by a Swedish astronomer Celsius. On this scale, the ice point of ice is 0°C and the boiling point at 100°C. Between these two intervals, the Fahrenheit scale is divided into 180 equal divisions and the centigrade scale into 100 equal divisions. Since both scales are linear, temperature can be easily converted from one to other, using the following equation:

$$\frac{^{\circ}\text{C}}{100} = \frac{^{\circ}\text{F} - 32}{180}$$

Kelvin and Rankine temperature scales: the Kelvin scale abbreviated °K was introduced about 1848 by Lord Kelvin. On the Kelvin scale the ice point is 273.15°K and the steam point is 373.15°K. The Kelvin scale like the centigrade scale is also divided into 100 equal divisions. The centigrade (°C) can be converted into Kelvin (°K) using the equation:

$$^{\circ}\text{K} = ^{\circ}\text{C} + 273.15$$

The Rankine scale: On the Rankine scale, which is abbreviated as °R, the ice point is 491.7°R and the steam point is 671.7°R. The Rankine scale, like the Fahrenheit scale is divided into 180 equal divisions. Temperatures in Fahrenheit (°F) can be converted into Rankine (°R) by using the equation:

$$^{\circ}\text{R} = ^{\circ}\text{F} + 459.69$$

Both Rankine and Kelvin temperature scales are called the absolute scales because they use absolute zero as one of their reference points.

The Reaumur scale: the Reaumur scale abbreviated as °R’ was introduced at about 1731. It assigns 0°R’ as the ice point and 80°R’ as the steam point. This often used in the alcohol industries

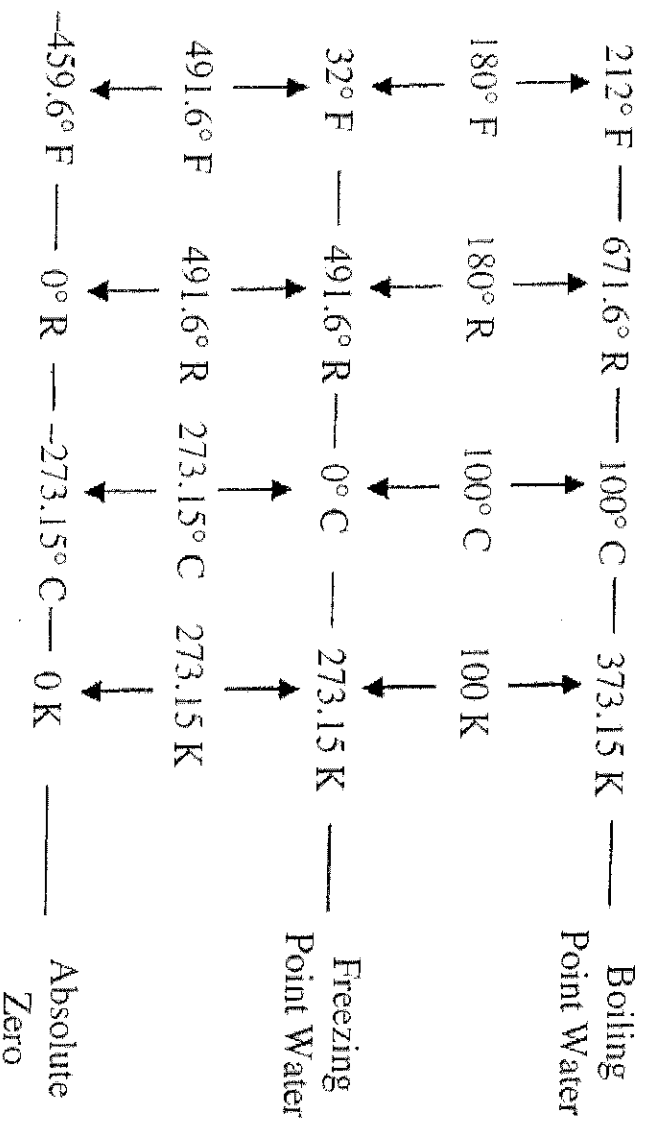


Fig 2: Comparison of various temperature sales

Methods of temperature measurement

The temperature measuring instruments are classified according to the nature of change produced in the testing body by the change of temperature. They may be classified as follows:

- Expansion thermometers
- Filled system thermometers
- Electrical temperature instruments
- Pyrometers.

1.3.1 Expansion thermometers: These thermometers are classified according to the substance which expands. They may be described under three headings as follows:

expansion of solids: a) bimetallic thermometers

expansion of liquids: a) liquid-in-glass thermometers

b) liquid-in-metal thermometers

iii) Expansion of gases: a) gas thermometers

Bimetallic thermometers: the expansion of solids is employed mainly in bimetallic elements by utilizing the difference in thermal expansion of bonded strip of two metals. It consists of two strip of metals invar and brass welded together; each strip made from a metal having a different co-efficient of thermal expansion. Whenever the welded strip is heated, the two metals will change length in accordance with their individual rates of thermal expansion. The two metals will expand to different length as the temperature rises. This forces the bimetallic strip to bend towards the side of low co-efficient of thermal expansion. If one end of the bimetallic strip is fixed so that it cannot move, the distance the other end moves is directly proportional to the square of the length of the strip as well as the total change in temperature, and inversely proportional to the thickness of the metal.

$$S \propto \frac{l^2 \Delta T}{t}$$

where; S – distance moved

l - Length of the metal strip

ΔT – total change in temperature

t - Thickness of metal

The movement of bimetallic strip is utilized to deflect a pointer over a graduated scale. The longer the length of the strip, the larger the deflection of the tip of the strip; since the distance is proportional to the square of the length. A longer can be contained in a relatively small space if the strip is wound in a spiral, a helix or a multi-helix form. If the bimetallic element is wound in a spiral form, the spiral coil is tightened with increase in temperature. As it coils, the counter post rotates clockwise, and thus a pointer attached to the post also moves on a calibrated temperature scale. (Fig 1.2). This type of temperature indicator is often used homes and offices to indicate ambient air temperature.

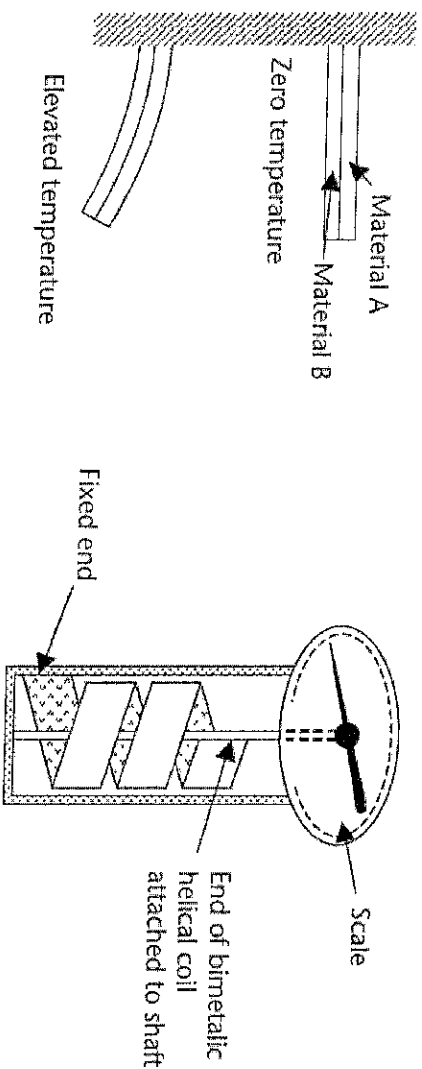


Fig 1.2 thermometer with spiral helical element.

The bimetal can also be used in the form of a helix to indicate temperature. This type of industrial bimetallic thermometer is shown in fig 1.3. It consists of a tightly wound bimetallic strip located inside the stem of the thermometer with one end fastened permanently to the outer casing. A strip is attached to a center post that extends from the stem to the center of the indicating dial. A pointer is attached to the center post. When the temperature surrounding the stem changes, the bimetal expands and the helical coil winds and unwinds which rotates the center post. This causes the pointer to move on the dial to indicate the measured temperature. Bimetallic thermometers are available for temperature range from -75 to 540°C.

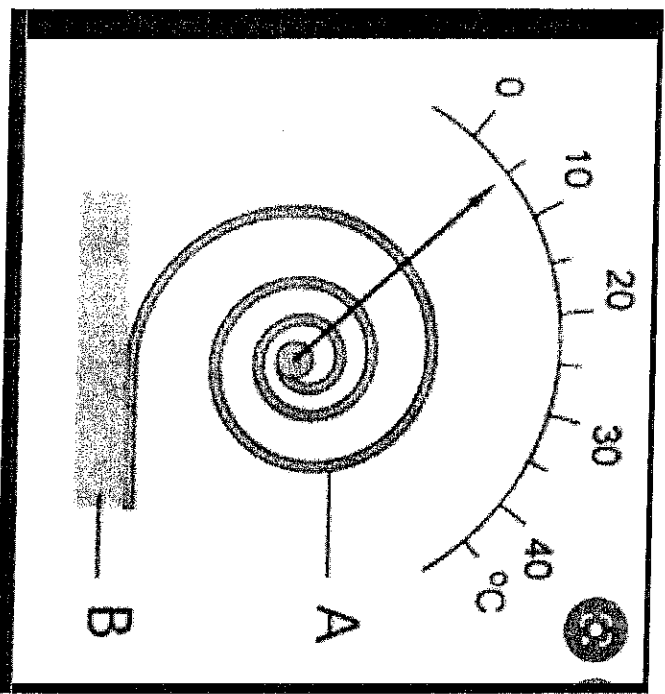


Fig 1.3 thermometer with spiral bimetallic element.

Advantages: the following are the advantages of bimetallic thermometers:

- low cost
- tough, not easily broken
- easily installed and maintained
- good accuracy relative to cost

fairly wide temperature range

Disadvantages: the following are the disadvantages of bimetallic thermometers:

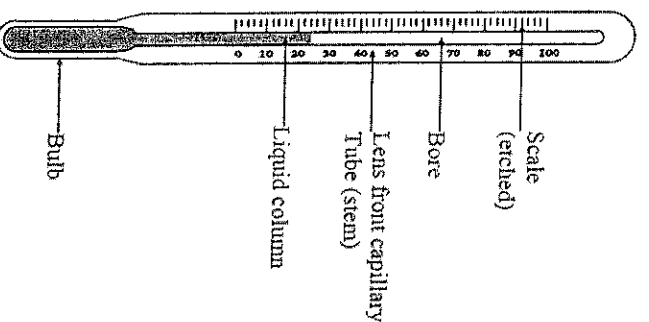
limitation to local mounting

availability of indicating type only

possibility of calibration changes due to rough handling

accuracy not as high as glass stem thermometers

Liquid-in-glass thermometers: the liquid in glass thermometer is one of the simplest temperature measuring devices, widely used in both laboratory and industry. Its operation is based on the fact that liquids expand as the temperature rises. In this type of thermometer, the expansion causes the liquid to rise in the tube, indicating the temperature. The simplest form of liquid in glass thermometer is shown in fig 1.4. It consists of a small bore glass tube with a thin-walled glass bulb at its lower end containing almost all the liquid (usually mercury). As heat is transferred into the mercury, the mercury expands, pushing the column of mercury higher in the



capillary above which indicates the temperature.

Fig 1.4 liquid in glass thermometer.

Liquid-in-glass thermometer is commonly used for temperature range of -120 to 320°C. But when mercury is used as liquid, it freezes at -39°C. Thus, for measuring very low temperature, alcohol is used as liquid. This type of thermometer cannot be used to measure temperatures higher than 600°C, because it permanent changes in the volume of the bulb, thus destroying the accuracy of the instrument.

The industrial mercury-in-glass thermometer is used in applications such as open tanks containing liquids, cooking kettles, steam lines, air ducts etc.

Advantages:

simple in construction

gives fairly accurate results

Disadvantages:

it is fragile

not easily adapted to automatic recording or transmission of temperature data
they can be difficult to read

Liquid-in-metal thermometer: the distinct disadvantages of liquid-glass-thermometers are overcome in liquid-in-metal thermometers. A liquid-in-metal is shown in fig 1.5. In which mercury has been used liquid and the metal is steel. This mercury-in-metal thermometer works on exactly the same principle as the liquid-in-glass thermometer. The glass bulb is replaced by a steel bulb and the glass capillary by one of stainless steel. Mercury is used as liquid in the system. As mercury in the system is not visible, a bourdon is used to measure the change in its volume. The bourdon, the capillary tube, and the bulb are completely filled with mercury, usually at higher pressure.

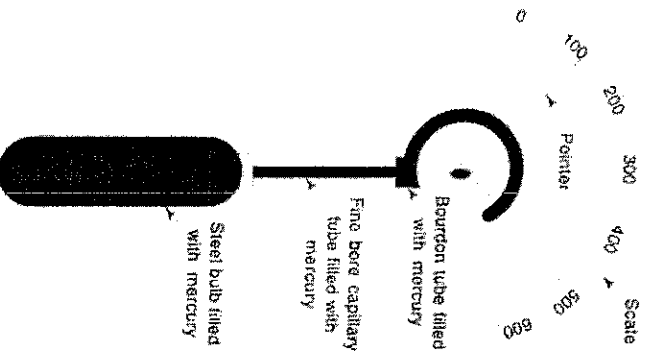


Fig 1.5 mercury in steel thermometer.

When the temperature to be measured rises, the mercury in the bulb expands more than the bulb so that some mercury is driven through the capillary tube to the bourdon tube. As the temperature continues to rise, increasing amount of mercury will be driven into the bourdon tube, causing it to bend. One end of the bourdon tube is fixed, while the motion of the other end is communicated to the pointer which moves on a calibrated temperature scale.

Generally, mercury is used as a liquid. But it has its limitations, particularly at the lower end of the temperature scale. For this and other reasons, other liquids are also used in place of mercury; these include xylene, alcohol, ether etc.

Gas thermometers: the operation of the gas thermometer is based on the ideal gas law which states that the volume of a gas increases with temperature, if the pressure is maintained constant; and the pressure increases with temperature if the volume is maintained constant. Therefore if a certain volume of inert gas is contained in a bulb, Capillary and bourdon tube, and must of the gas in the bulb, then the pressure indicated by the bourdon tube may be calibrated in terms of temperature of the bulb.

Nitrogen is the favorite fill for the gas filled thermometer because it is almost inert and inexpensive. It does react somewhat with the steel at temperatures exceeding 427°C , and it does act less like a perfect gas at extremely low temperatures. Under these conditions, helium should be used.

An advantage of the gas thermometer is that the gas in the bulb has a lower thermal capacity than a similar quantity of liquid, so that the response to temperature changes will be more rapid than for a liquid-filled system with a bulb the same size and shape. It has a temperature range of -200 to 1500°C depending on the gas used.

1.3.2 Filled system thermometers: These consist of a bourdon tube, a capillary tube and a thermometer bulb all interconnected as shown in fig 1.6. The entire system is sealed after filling with the appropriate liquid (known as filling liquid) at a pressure at the normal ambient temperature. The common liquids used are mercury, ethyl alcohol, xylene and toluene. When in use the thermometer bulb is inserted inside the substance to be measured. This causes the filling liquid inside the bulb to heat or cool until its temperature matches that of the substance. This change in temperature causes the filling liquid to expand or contract and thus the bourdon tube moves. With increase in temperature the liquid expands and this forces the bourdon tube to uncoil. With decrease in temperature the liquid contracts and it forces the bourdon tube to coil more tightly. The movement of the bourdon tube may be used to drive a pointer for indicating temperature.

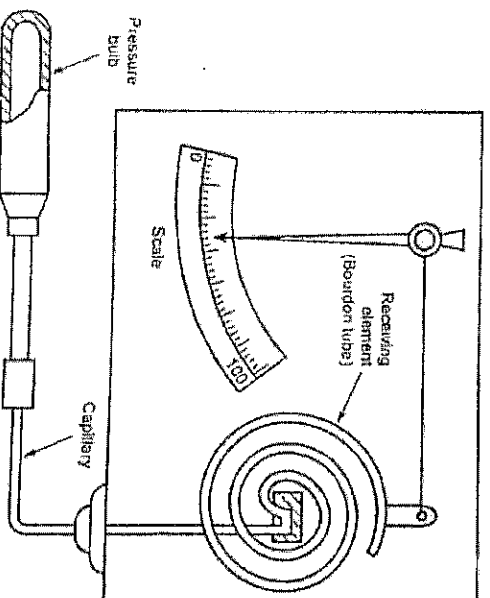


Fig 1.6 filled system thermometer

Basically, there are four types of filled system thermometers:

Gas filled thermometers (already discussed)

Liquid filled thermometers

Mercury filled thermometers

Vapour pressure thermometers

Liquid-filled thermometers: these thermometers are completely filled with liquids (other than mercury) and operate on the principle of liquid expansion with increase in temperature. The filling is usually an inert hydrocarbon such as xylene (C_8H_{10}) which has a co-efficient of expansion six times that of mercury and makes smaller bulb possible. Other liquids (even water) are sometimes used. The criterion is that the pressure inside the system must be greater than the vapour pressure of the liquid to prevent bubbles of vapour from forming. Also, the liquid should not be allowed to solidify even in storage; otherwise, the calibration may be affected.

Vapour pressure thermometers: in this system the bulb is partially filled with liquid, while the capillary tube and the bourdon are filled with vapour. In this system some of the liquid vaporizes during operation.

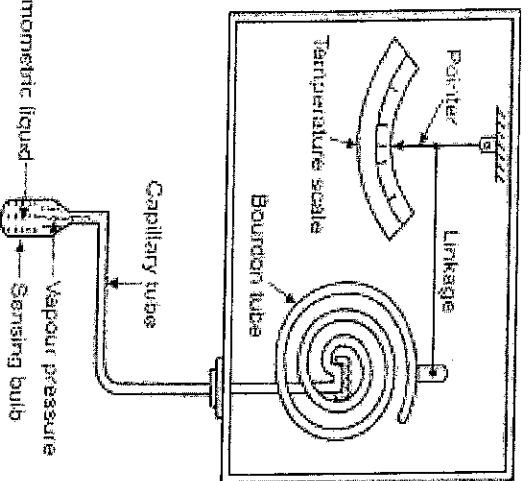


Fig 1.7 vapour pressure thermometer.

A vapour pressure thermometer is shown in fig1.7. The liquid in the system boils and vaporizes during operation which creates a gas or vapour inside the capillary and bourdon tube. The liquid inside the bulb continues to boil until the pressure in the system equals the vapour pressure of the boiling liquid. At this point the liquid stops boiling unless its temperature increases. When the temperature of the substance surrounding the bulb drops, the liquid and the vapour inside the bulb also cool, causing some of the vapour to condense. As the vapour condenses, the pressure inside the system decreases. Due to this change in pressure, the bourdon tube uncoils as the pressure increases and coils as tightly as it decreases. This movement of the bourdon tube may be connected to a pointer, or to a pen on a strip chart recorder, or to a transmitter to indicate temperature.

Advantages: the advantages of filled system thermometers are:

Rugged construction

Low maintenance requirement

Absence of the need for electric power and self-sufficiency.

The point of display can be located at a considerable distance from the point of measurement.

Satisfactory time response sensitivity and accuracy for most industrial application.

Low cost.

Delivers enough power to drive not only a pointer or a recording pen but also a controller mechanism

Three (or more) separate systems can be put in a single instrument case.

The capillary allows considerable separation between the point of measurement and the point of temperature indication. Usually up to 30m is obtainable.

Disadvantages: the disadvantages of filled system thermometers are:

Need for a large bulb for the sake of accuracy

The entire system has to be replaced in case of failure.

Much lower accuracy, sensitivity, and temperature span compared to electrical temperature instrument.

Limited maximum temperature compared to some electrical measuring system.

Thermal system, being under pressure, cannot be broken without destroying the calibration.

Separation of measuring element and bourdon of more than 75m generally is not recommended, it is frequently more economical to employ transmitters for signal transmission beyond 30 m.

1.3.3 Electrical temperature instruments: there are three types of electrical measuring instruments which are generally used in industries:

Resistance thermometer

Thermocouple

Thermistor

Resistance thermometer: the resistance of certain metals changes with change in temperature. Resistance thermometer utilizes this characteristic. With the increase in temperature, the electrical resistance of certain metals increases in direct proportion to the rise in temperature. Therefore, if the electrical resistance of a wire of known and calibrated material is measured, the temperature of the wire can be determined. Platinum, copper, and nickel are generally used in resistance thermometers.

In this type of thermometer, a temperature sensitive element is fabricated and inserted in the medium whose temperature is to be measured, and is connected by leads to a Wheatstone bridge as shown in fig 1.8a. The bridge consists of a sensing element of resistance X having high temperature co-efficient and resistance A, B, and C whose ohmic value do not alter with change in temperature. LR1 and LR2 are the lead wires resistances of the sensing element. The principle of the Wheatstone bridge states that in balanced condition (when no current flow through the galvanometer), the ratio of the resistance

$$\frac{A}{R} = \frac{(X + LR_1 + LR_2)}{C}$$

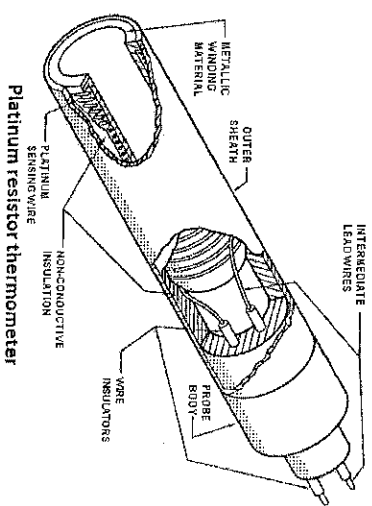
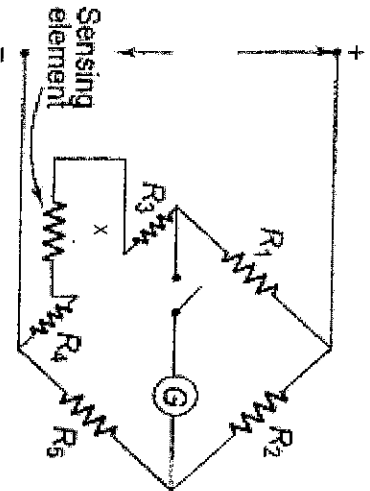


Fig 1.8a resistance thermometer bridge circuit. b a resistance thermometer

Now, when the resistance X changes, the Wheatstone bridge becomes unbalanced and the galvanometer will give deflection which can be calibrated to give a suitable temperature scale.

Resistance elements are generally long, spring like wires enclosed in a metal sheath. The construction of a practical resistance thermometer is shown in fig 1.8b. The resistance element is surrounded by porcelains which prevents short circuit between the wires and the metal sheath. Two leads are attached to each side of the platinum wire. When this instrument is placed in liquid or gas medium whose temperature is to be measured, the sheath quickly reaches the temperature of the medium. This change in temperature causes the platinum wire to inside the sheath to heat or cool, resulting in proportional change in the wire's resistance. This can be calibrated to indicate the temperature.

Conductors used for resistance thermometer are platinum, balco (70% nickel and 30% iron) and copper, listed in their decreasing order of temperature range capabilities. A comparison chart of various resistance temperature detectors RTD or sensors is given in table 1.2.

Advantages: the following are the advantages of resistance thermometers:

- High accuracy of measurement.
- Wide temperature range (from -200 to 650°C).
- Small size.
- Fast response.
- Good reproducibility.
- Stable and accurate performance over many years.
- Temperature compensation not required.
- Disadvantages:** the following are the disadvantages of resistance thermometers:
 - High cost.
 - Need for a bridge circuit and power supply.
 - Inaccuracy resulting from current flowing through the bridge circuit thereby heating the resistance element.
 - Mechanical abuse or vibration.
 - Larger bulb size than thermocouples.

Thermocouples: the working principle of a thermocouple depends on the thermo-electric effect. If two dissimilar metals are joined together so as to form a closed circuit, there will be two junctions where they meet each other. If one of these junctions is heated, then a current flow in the circuit which is detected by a galvanometer. The amount of current produced depends on the difference in temperature of the two junctions and the characteristics of the two metals. This was first observed by Seebeck in 1821 and is known as Seebeck effect.

Instruments which record the variations in current flow are calibrated in terms of temperature and are known as thermocouple pyrometers. Fig 1.9a shows a thermocouple made from two different kinds of metals. These wires are joint at the ends which form the two junctions, a measuring junction and a reference junction. Heating the measuring junction produces a voltage greater than the voltage at the reference junction. The difference between the two voltages is measured and the voltage reading is converted into its corresponding temperatures. This conversion table is generally supplied by the thermocouple manufacturers. There are various types of which are used in the industries, viz, copper constantan, iron constantan, chromel alumnel, platinum rhodium, and chromel constantan.

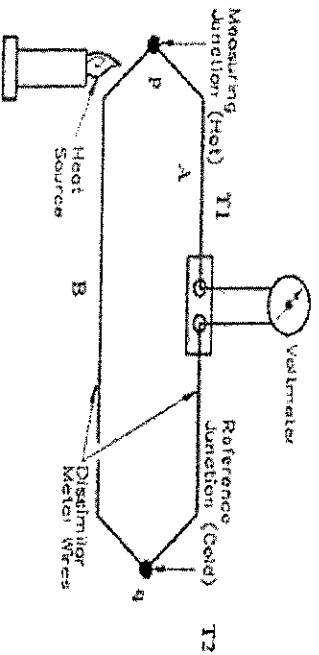


Fig1.9a thermocouple.

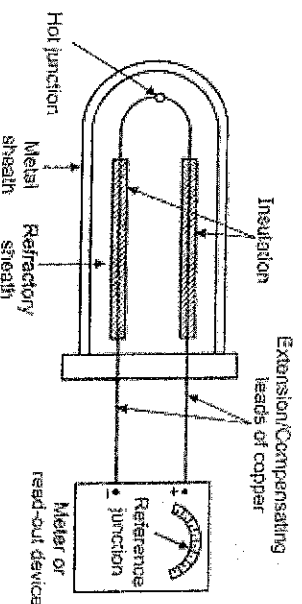


Fig 1.9b a practical thermocouple

A practical thermocouple used in industries, is shown in fig 1.9b. It consists of a protective well and head across the measuring junction to avoid damage in pressurized systems or when measuring corrosive fluid. The measuring junction is connected to the reference junction and the voltmeter with extension wires (also known as compensating leads). The characteristics of the compensating wires must be similar to those of the thermocouples wires it joins; otherwise, the voltage generated at the connection will be large enough to affect the instruments' accuracy. A

volt meter reads the difference between the measuring and the reference junctions' voltages which can be calibrated to give the temperature reading.

Thermocouples are not limited to single point measurement. They can be connected in parallel to provide the average temperature in a system. They can also be used to measure the difference between two temperatures. Based on possible combination of metals, there are a large number of thermocouples available. Table 1.3 shows the comparison between different thermocouples.

Advantages: the following are the advantages of thermocouples:

Rugged construction.

Inexpensive.

Simpler to use than resistance thermometers.

Absence for the need for bridge circuit.

Extremely wide temperature ranges from -270 to 2800°C.

Wide variety of design to both standard and special application.

Electrical output adaptable to a number of readout and/or control devices.

High response speed compared to filled system thermometers.

Good accuracy.

Calibration checking made easy.

Long transmission distances.

Good reproducibility.

Disadvantages: the following are the disadvantages of thermocouples:

Limited use in temperature spans less than about 33°C because of the relatively small change in junction voltages with temperature.

Extension leads must be housed in metal conduit, as low junction voltage can cause the device to pick up stray electrical signals.

The need to hold reference junction temperature constant or compensation for any deviation.

Non linearity of voltage-temperature relationship.

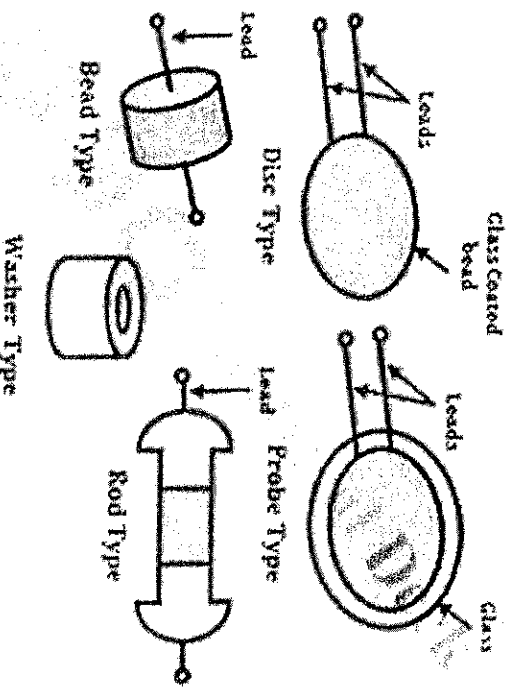
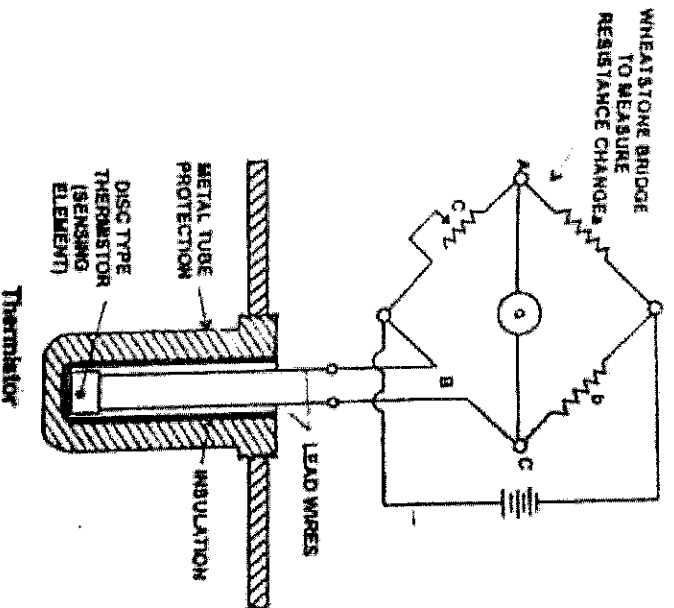
Temperature spans not being as narrow as filled system or resistance thermometers.

Require much of amplification for many applications.

Need for expensive accessories for control application.

Thermistors: Thermistors (also known as “thermally sensitive resistors”) are semiconductors, made from a specific mixture of pure oxides of nickel, manganese, copper, cobalt, iron magnesium, titanium and other metals sintered at temperatures above 982°C.

Thermistors are available in a number of configurations, most familiar is the bead type usually glass coated. They can also be made into washers, discs, or rods, and can also be encapsulated in plastic, cemented, soldered in bolt, encased in glass tubes, needles, or a variety of other forms.



To measure temperature with a Thermistor, it is placed in the environment whose temperature is to be measured. As the temperature of the substance or environment increases, the resistance of the Thermistor increases and vice versa. This change in Thermistor resistance can be detected which will be the measured of the temperature of the substance. Generally, the Thermistor is placed as one leg of a Wheatstone bridge as shown in fig 1.10. At balance condition, when there is no change in temperature, the galvanometer indicates zero. As the temperature increases or decreases, the resistance of the thermistor also increases or decreases due to which the Wheatstone bridge circuit becomes unbalanced. Thus an electric current flows through the galvanometer whose deflection can be calibrated to measure temperature.

Advantages: the following are the advantages of thermistors:

- Small size and fast response.

- Suitability for narrow spans.

- Low cost.

- Sensor exhibiting a greater sensitivity as the temperature drops, when temperature co-efficient is negative.

- Resistance is a function of absolute temperature, so cold junction compensation not necessary.

- Due to large sensor (thermistor) resistance, the contact or Lead wire resistance considered as negligible.

- Stability of the instrument increases with age. They are adaptable to various electrical readout.

Disadvantages: the following are the disadvantages of Thermistors:

- Nonlinear temperature versus resistance curve.

- Unsuitable for wide temperature spans.

- Problem due to interchangeability of individual element.

- Doubtful stability at higher temperatures.

- Being limited for process application.

- Too much resistance requiring shielded power lines, filter, or DC voltage.

1.3.4 Pyrometers: all the temperature measuring discussed earlier require physical contact of the thermometer with the body whose temperature is to be measured. At high temperatures (above 1400), the thermometer may melt due to direct physical contact. To solve these problems a non-contact method of temperature sensing is used. Also for bodies that are moving, a non contacting temperature sensing is most convenient.

Pyrometry is a technique for measuring temperature without physical contact. It depends upon the temperature of the hot body and the electromagnetic radiation emitted by the body. When a body is heated, it emits thermal energy known as heat radiation. A black matt surface (or a black body) is a very good absorber of heat and, also, a very good emitter of such radiation when heated. Pyrometry is a process of determining a body's temperature by measuring its electromagnetic radiation. There are two types of pyrometers generally used in industries:

Radiation pyrometers.

Optical pyrometers.

Radiation pyrometers: operation of radiation pyrometer is based upon the radiant energy emitted by the hot body. It consists of a lens to focus the radiated energy from the body, whose temperature is required, onto a detector or receiving element (fig 1.11). This receiving element may have a variety of forms such as resistance thermometer, or a thermocouple, or thermopile. A thermopile consists of several thermocouples connected in series. A temperature indicator, recorder, or controller is attached with the receiving element to record the temperature.

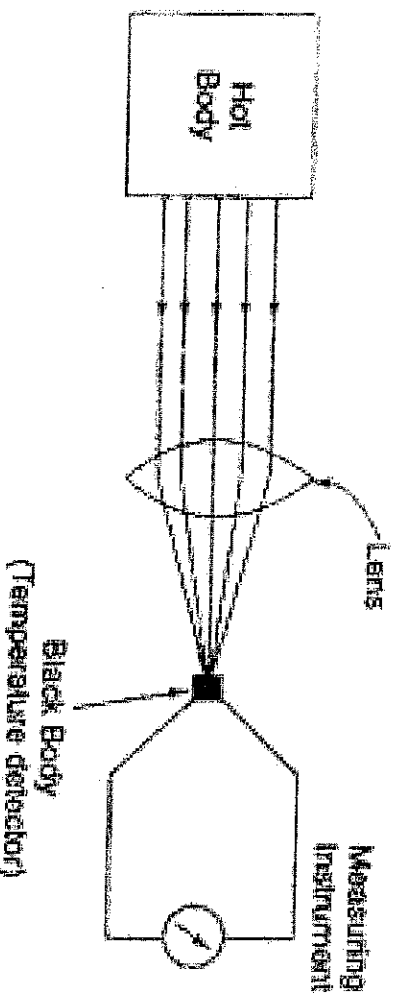


Fig 1.11 a typical radiation pyrometer.

When the total energy radiated by a hot body, whose temperature is to be measured, enters the pyrometer, it is focused by the lens on to the detector. The detector is a thermopile whose measuring junctions are attached to a blackened disk. The disk absorbs energy when the pyrometer is focused on a hot body, and its temperature rises. The reference junction of the thermopile is attached to the pyrometer case. The difference in temperature between the measuring junction attached to the disc and the reference junction attached to the case generates a voltage that is directly related to the temperature of the blackened disk, which is indicated by the recording instrument.

Advantages: the following are the advantages of radiation pyrometers:

Ability to measure high temperature.

No need of contact with target of measurement.

Fast response speed.

High output and moderate cost.

Disadvantages: the following are the disadvantages of radiation pyrometers:

A non-linear scale.

Possible errors due to presence of intervening gases or vapours that absorb radiating frequencies.
Emissivity of target material affecting measurement.

Applications: the following are the applications of radiation pyrometers:

They are used for the temperature above the practical operating range of thermocouples.

They can be used in environments which contaminate or limit the life of thermocouples.

For fast moving targets.

They are used for targets not easily accessible, such as furnace interior.

They are used for targets which would be damaged by contact with primary elements like thermocouples and resistance thermometers.

Optical pyrometers: optical pyrometers provide an accurate method of measuring temperature between 600 and 3000°C, and are very useful for checking and calibrating radiation pyrometers. But they are not suitable for recording and controlling temperatures.

1.5 calibrations of thermometers: to calibrate an instrument, its indicated temperature or electrical output is compared to a standard device at a number of known points. By estimating the values between the known points, output of the instrument can be determined over a useful temperature range.

2.0 Techniques of pressure measurement

Pressure: pressure is defined as the amount of force applied to a surface or distributed over it and is measured as force per unit area. This force must act normally to the surface. If it acts at a slant, only part of it which acts is used to calculate the pressure. The part of the force which acts parallel to the surface does not contribute to pressure.

Unit of pressure: in metric (SI unit) pressure is measured in Newton per square meter.

One Newton per square meter = one Pascal

$1 \text{ N/m}^2 = 1 \text{ Pa}$

One atmosphere (1 atm) = 101325 N/m² or

1 atm = 101.325 KN/m² = 101.325 KPa

In British unit, pressure is measured in pounds per square inch (psi)

1 atm = 14.696 psi = 101.325 KPa

1 psi = 6.895 KPa

1 bar = 10⁵ N/m²

Low pressure units:

1 millibar = 100 dyne/cm² = 14.5 x 10⁻³ psi

1 micron = 10-6 m Hg = 19.34 x 10⁻⁶ psi

1 torr = 1 mm Hg = 1000 microns = 19.34 x 10⁻³ psi

A pressure conversion table is shown below to convert from one unit to the other:

Table 2.1 pressure conversion table.

TO CONVERT FROM	TO TORR	MULTIPLY BY
Mm Hg		1
Inches of Hg		25.4
Inches of water		1.87
Feet of water		22.39
Atmosphere (bar)		760
Psia		51.7
KPa		7.5
Microns		0.001
TO CONVERT TO	FROM TORR	DIVIDE BY

Types of pressure: when pressure is measured, it is usually desired to read it in terms of either gauge pressure, absolute pressure, or vacuum (differential) pressure.

Gauge pressure: most liquid pressure gauges use atmospheric pressure (14.7 psi) as a zero point, I e they indicate a pressure of zero psi at the surface of the liquid even though the pressure is actually 14.7 psi (101.325 KPa).

A gauge that indicates zero at atmospheric pressure measures the difference between the actual pressure and atmospheric pressure. This difference is called “gauge pressure”. It is written as psig (pound per square inch gauge).

Absolute pressure: absolute pressure is actual total pressure (including atmospheric pressure) acting on a surface. It is written as psia (pounds per square inch absolute).

Vacuum or Differential pressure: gauges that indicate gauge pressure may be designed to indicate pressure below zero. Such a gauge is called a “vacuum gauge”. Gauges that indicate absolute pressure cannot indicate pressures below zero, because zero is a perfect vacuum.

In a differential pressure measurement, the gauge pressure is the difference between the absolute pressure of the fluid and the atmospheric pressure.

The relationship between absolute, gauge and atmospheric pressure is shown in fig 2.1.

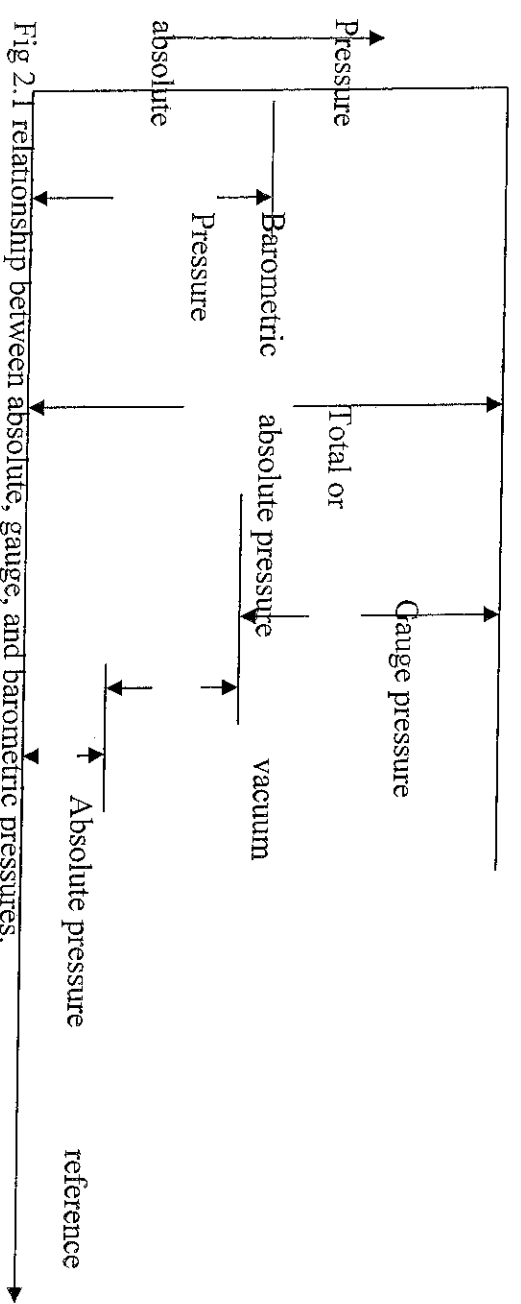


Fig 2.1 relationship between absolute, gauge, and barometric pressures.

pressure measuring instruments: most pressure measuring instruments measure a difference between two pressure one usually that of the atmosphere. The different types of pressure measurement are listed below:

Manometer method.

Elastic pressure transducers.

Electrical pressure transducers.

2.4.1 Manometers: the manometer is the simplest instrument used for measuring gauge pressure (low range pressure), by balancing the pressure against the weight of a liquid column. The action of all manometers depends on the effect of the pressure exerted by a fluid at a depth. The different types of manometers are discussed below:

The U-tube manometer: the U-tube is the simplest form of manometer and is used for experimental work in the laboratories. By suitable choice of liquids, wide range of temperature can be recorded. They are usually used as indicators of fluid pressure, but rarely for control application, unless coupled to laser, ultrasonic or other means of level measurement device.

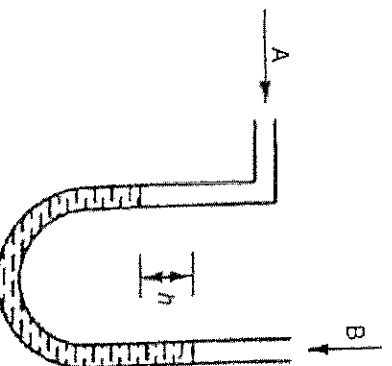


Fig 2.2 U-tube manometer

Construction: it consists of a transparent (glass) tube constructed in the form of an elongated U and partially filled with a liquid, most commonly water and mercury. Water and mercury are

used because their specific weights at various temperatures are known exactly and they do not stick to the tube. One end of the tube is connected to one pressure tap and the other end is connected to the other pressure tap, or it may be left open to the atmosphere. The U-tube manometer is shown in fig 2.2

Working: when there is a pressure difference between the two ends of the tube, the liquid level goes down on one side of the tube and up on the other side. The difference in liquid levels from one side to the other indicates the difference in pressure. From fig 2.2 the difference in pressure ($P_1 - P_2$) is obtained by the relation:

$$(P_1 - P_2) = (\rho - \rho_1)(h_1 - h_2)g \text{ or}$$

$$P = (\rho - \rho_1)gh.$$

Where: ρ = density of fluid in U-tube

ρ_1 = density of fluid whose pressure is being measured

$h = (h_1 - h_2)$, difference in fluid level

g = acceleration due to gravity

When a manometer is used to measure low pressures then water is used as the liquid, and when the manometer is used to measure high pressures, the mercury is used as the liquid.

Limitations: in the U-tube manometer, the application of pressure causes the liquid in one leg to go down while the other leg goes up, so there is no fixed reference. This tends to make measurement of the height more difficult than it would be if one surface could be maintained at some fixed level.

Well-type manometer: this type of manometer is widely used because of the convenience in its requiring the reading of only a single leg. It consists of a very large diameter vessel (well) connected on one side to a very small-sized tube. Thus, the zero level moves little when pressure is applied. Even this small error is compensated by suitably distorting the length of the scale. However, such an arrangement is sensitive to non-uniformity of the tube cross-sectional area and is thus considered to be somewhat less accurate.

In a single-leg instrument, high accuracy is achieved by setting the zero level of the well at the zero level of the scale before each reading is taken. Fig 2.3 shows a well-type manometer.

Barometer: since manometers inherently measure the pressure difference between the two ends of the liquid column, if one end is at zero absolute pressure, then the difference in height of the liquid from the zero reference indicates the absolute pressure.

A barometer is a well type absolute pressure gauge whose pressure range is from absolute zero to atmospheric pressure. The readings are generally in millimeters of mercury (mm Hg). With a barometer high vacuum are not measured. Fig 2.4 shows a barometer.

Inclined manometer: the inclined manometer or slant manometer is an enlarge leg manometer with its measuring leg inclined to the vertical axis by some angle. The angle of inclination is of the order of 10° . The inclination is done to expand the scale and thereby increase sensitivity.

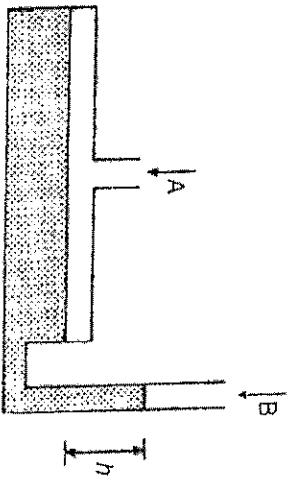


Fig 2.3 well type manometer

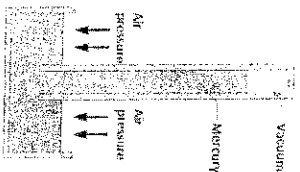


fig2.4 a barometer

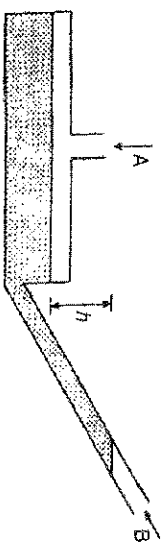


fig2.5 inclined manometer

The inclined manometer is used to measure very small pressure difference. The manometer is tipped so that so that the liquid moves a longer distance through the tube as it rises. Fig 2.5 shows an inclined manometer. The distance y that the liquid moves through the tube is greater than the distance x that the liquid rises.

For accurate measurement of extremely small pressure difference, a micro manometer is used.

Fluids for manometers: the most common fluids used for manometers are water, red oil and mercury. Water is the best fluid if air is the fluid to be measured and the glass length is within reasonable limits.

Mercury is used when the measured fluid mixes with manometer fluid or a very long tube is necessary, while red oil is used as the indicating fluid when it is required to read the manometer from a distant place. It is easy to read and does not combine chemically with some fluid in a dry system.

Advantages: the following are the advantages of manometer:

Simple and time proven.

High accuracy and sensitivity.

The availability of wide range of filling fluids of varying specific gravity.

Reasonable cost.

Their suitability for low pressure and low differential pressure application.

Disadvantages: the following are the disadvantages of manometers:

Large and bulky.

Need for leveling.

Lack of portability.

Measured fluid must be compatible with manometer fluid.

No over-range protection.

Condensation may present problems.

2.4.2 Elastic pressure transducers: These types of pressure transducers use elastic primary sensing elements such as the bourdon tube, bellows and diaphragm. They are sometimes referred to as indirect pressure transducers.

Bourdon tube: the bourdon tube is the most frequently used pressure gauge because of its simplicity and rugged construction. The bourdon tube has an elliptical cross section and can be bent into various configurations: C-type, spiral, helical, or twisted. It covers ranges from 0-15 psig to 0-100,000 psig, as well as vacuum from 0 to 30 inches of mercury.

Construction and working of a C-type bourdon tube: a C-type bourdon tube consists of a long thin walled cylinder of non-circular cross-section, sealed at one end, made from materials such as phosphor bronze, steel and beryllium copper, and attached by a light line work to the mechanism which operates the pointer. The other end of the tube is fixed and is open for the application of the pressure which is to be measured.

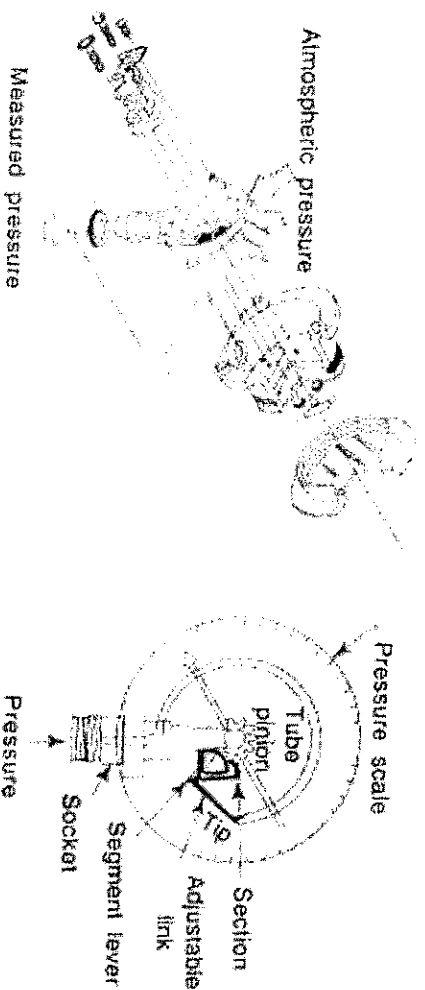


Fig 2.6 C-type bourdon tube

As fluid under pressure enters the bourdon tube, it tries to change the section of the tube from oval to circular, and this tends to straighten out the tube. The resulting movement of the free end of the tube causes the pointer to move over the scale. The tip of the bourdon tube is connected to a segmental lever through an adjustable length link. The pressure inside the tube must be equal to the atmospheric pressure outside the tube. For adequate reliability, materials for bourdon tubes must have good elastic or spring characteristics.

Advantages: the following are the advantages of bourdon tubes:

- Low cost.
- Simple construction.
- Time-tested in applications.
- Availability in a wide variety of ranges, including very high ranges.
- Adaptability to transducer design for electronic instruments.
- High accuracy, especially in relation to cost.
- Disadvantages:** the following are the disadvantages of bourdon tubes:
 - Low spring gradient.
 - Susceptibility to shock and vibration.
 - Susceptibility to hysteresis.

Diaphragm pressure transducers: diaphragms are widely used for pressure (gauge pressure) and draft measurements, particularly in very low ranges. They can detect a pressure difference even in the range of 0 to 4 mm. the diaphragm can be in the form of flat, corrugated or dished plates and the choice depends on the strength and amount of deflection desired. In high precision instruments the diaphragms are generally used in pair, back-to-back, to form an elastic capsule.

With suitable modification, diaphragm elements are made to cause in electric circuit, thus converting pressure movement to electrical signal which can be transmitted to an indicating or recording system. Two types of diaphragms are generally used:

Metallic diaphragm gauge.

Slack diaphragm gauge.

Metallic diaphragm gauge: it consists of a thin flexible diaphragm made of materials such as brass or bronze. A pointer is attached to the diaphragm as shown in fig 2.7. The force of pressure against the effective area of the causes a deflection of the diaphragm. In some the deflection of the diaphragm is opposed by the spring qualities of the diaphragm and in other cases a spring is added to limit the deflection of the diaphragm. The motion of the diaphragm operates an

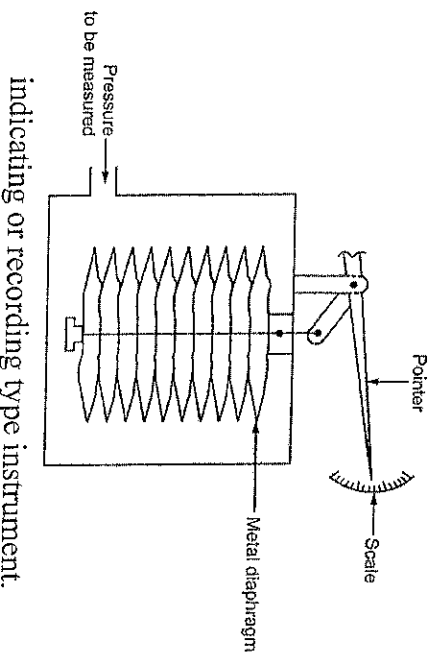


Fig 2.7 a metallic diaphragm

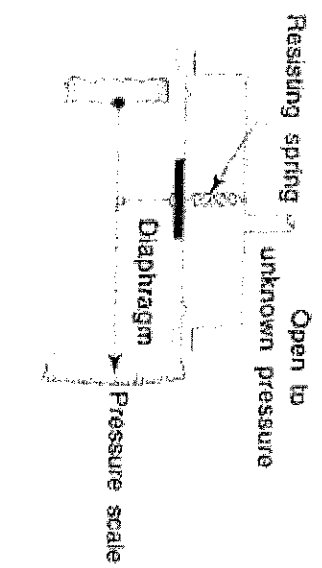


fig 2.8 slack-type diaphragm

This type of gauge is capable of working in any position and is portable, and therefore well adapted for use or installation in moving equipment such as aircrafts.

Slack diaphragm gauge: it is more difficult to measure pressure below atmospheric because the changes are small. Pressures in this range can be measured with a slight modification of the diaphragm. A diaphragm with a large area produces a large change in force from a small change in pressure. Similarly, making the diaphragm slack rather than tight allows it to move a large distance in response to a small change in pressure. A slack diaphragm can be made of rubber or other flexible materials. Fig 2.8 Shows a slack diaphragm gauge.

A slack diaphragm gauge with weak spring and a large area can be used over pressure ranges as low as 0.01 – 0.40 mm Hg.

Advantages: the following are the advantages of diaphragm elements:

- Moderate cost.
- High over range characteristics.
- Adaptability to absolute and differential pressure measurement.
- Good linearity.
- Available in different material for corrosion resistance.
- Small size.
- Adaptability to slurry services.

Disadvantages: the following are the disadvantages of diaphragm elements:

- Lack of good vibration or shock resistance.
- Difficult to repair.
- Limited to relatively low pressures.

Bellows: bellow-type gauges are used for of absolute pressures (normal as well as low pressure). It is somewhat more sensitive than bourdon-tube gauges. It is generally used for the range down to 155.1 mm Hg (3 psi). It may be used for even lower pressure up to 40 mm Hg by making the bellows large enough.

Primary conversion element, e.g. resistance or voltage.

Secondary conversion element.

The following are the commonly used electrical transducers:

Strain gauge pressure transducer: this is a passive type resistance transducer whose electrical resistance changes when mechanically stretched or compressed (strained) due to physical effects. It can be attached to a pressure sensing diaphragm so that when the diaphragm flexes due to the process pressure applied to it, the strain gauge stretches or compresses. This deformation of the strain gauge causes a variation in its length and cross-sectional area due to which its resistance also changes (fig2.10a)

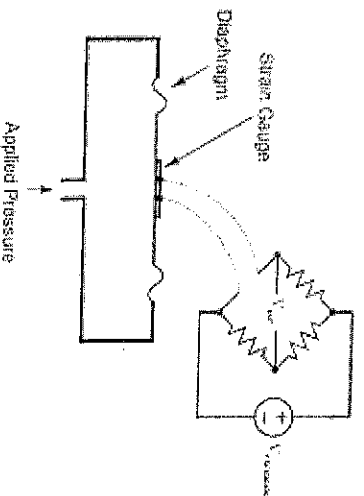


Fig 2.10a strain gauge transducer with

Diaphragm element

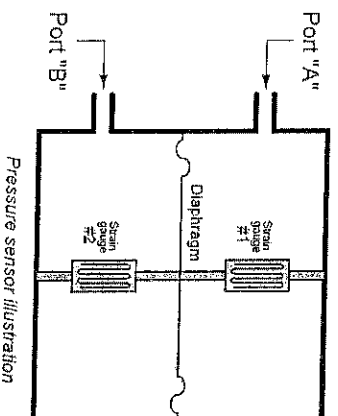
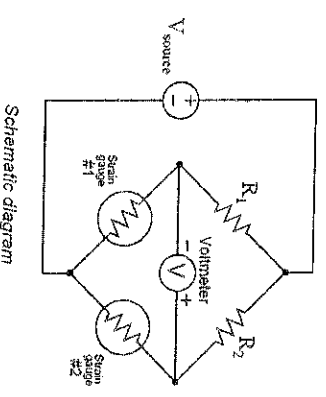


fig 2.10b strain gauge bridge circuit



The resistance change of a strain gauge is usually converted into voltage by connecting one, two, or four similar gauges, as of a Wheatstone bridge (known as Strain Gauge Bridge), and applying excitation to the bridge. The bridge output voltage is then a measure of the pressure sensed by the strain gauge.

Construction and working: it is usually constructed in form of a bridge circuit with four strain gauges, Rsg 1, Rsg 2, Rsg3 and Rsg 4. Two strain gauges, Rsg 1 and Rsg 4, are mounted so that increasing pressure increases their resistance. Strain gauges Rsg 2 and Rsg 3 are mounted so that increasing pressure decreases their resistance.

At balance when there is no pressure, no current flows through the galvanometer G, and hence there will be no deflection of the galvanometer. As soon as pressure is applied, the strain gauge stretches or compresses accordingly and the bridge circuit is unbalanced due to the change in resistance of the strain gauges. These changes affect the output of the bridge circuit, which indicates a change in measured pressure. Now, this change in output voltage may be calibrated for the pressure change.

Advantages: the following are the advantages of strain gauge pressure transducers:

Small and easy to install.

Have good accuracy.

Availability for wide range of measurements (from vacuum to 200,000 psig).

Good stability.

High output signal strength.

High over range capacity.

Simple to maintain.

Contain no moving parts.

Good shock and vibration characteristics.

Readily adaptable to electronic systems.

Fast speed of response.

Disadvantages: the following are the disadvantages of strain gauge pressure transducers:

Moderate to high cost.

Electrical readout necessary.

Requires constant voltage supply.

Temperature compensation required due to problems presented by temperature variations.

Potentiometric pressure transducers: in this type of pressure transducer, a potentiometer (basically a variable resistance) which is made by winding resistance wire round an insulated cylinder. A movable contact, called the wiper, slides along the cylinder, touching the wire at one point on each turn. The position of the wiper determines how much wire, and therefore, how much resistance is between the end of the wire and the wiper. A mechanical linkage from a pressure sensing device (such as bellows, diaphragm, or bourdon tube) controls the position of the wiper on the potentiometer. The position of the wiper determines the resistance of the potentiometer which in turn determines the pressure. Fig 2.11 Shows a Potentiometric pressure transducer in which the sensing element is a bourdon tube.

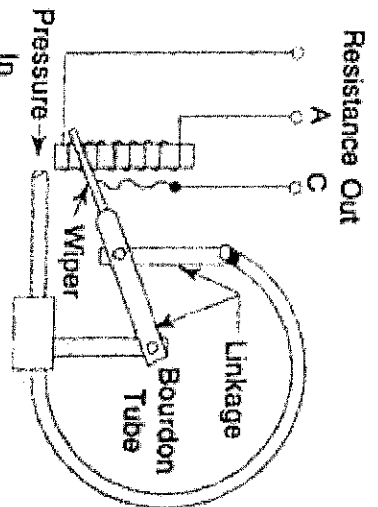


Fig 2.11 Potentiometric pressure transducer

Advantages: Potentiometric pressure transducers are widely used in industries for the following reasons:

The resistance can be easily converted into a standard voltage or current signals.

These pressure transducers are simpler and less expensive than other types of pressure transducers.

They are easy to design to meet special requirements in specific applications.

Disadvantages: the following are the disadvantages of Potentiometric pressure transducers:

They have a finite resolution, i.e. the wiper does not move continuously along the wire.

Wear occurs as the wiper moves back and forth across the wire.

Capacitive pressure transducers: the principle of operation of capacitive pressure transducer is based upon the familiar capacitance equation of the parallel plate capacitor, i.e.

$$C = \frac{\epsilon_o \epsilon_r A}{d}$$

Where: C = the capacitance of a capacitor in farad

A = area of each plate in mm

D = distance between the two plates in m

$$\epsilon_0 = 8.864 \times 10^{-12} \text{ farad/m}^2$$

ϵ_r = dielectric constant (relative permittivity)

Thus the capacitance of a capacitor varies when,

The area A of the plate is changed

The distance d between the plates is changed

The dielectric constant ϵ_r is changed

Now, from the above equation, the capacitance of a parallel plate capacitor is inversely proportional to the spacing between the plates

$$\text{(i.e. } C \propto 1/d\text{)}$$

This is made use of in the capacitive pressure transducer for detecting pressure changes.

Construction and working: it consists of a fixed plate and a movable plate which is free to move as the pressure applied changes. According to the change in pressure the movable plate also changes its position, due to which the distance d is changed. With increase in pressure, the distance d becomes less, due to which the capacitance C is increased and vice versa. This change in capacitance can be calibrated to measure change in pressure.

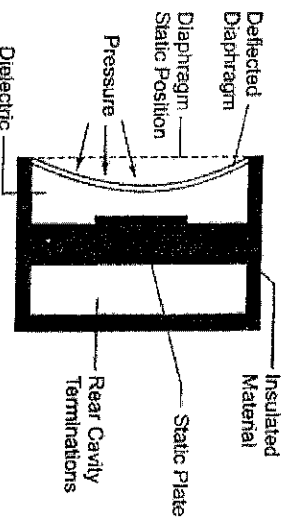


Fig 2.12a capacitive pressure transducer

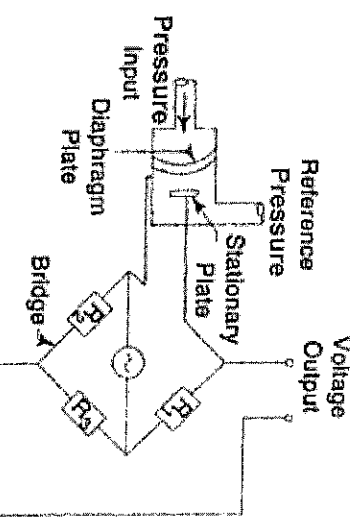


fig 2.12b diaphragm plate capacitive

In place of a movable plate, a diaphragm may be used, as shown in fig 2.12b, which expands and contracts due to change in pressure. The diaphragm plate acts as the movable plate of a capacitor. A fixed plate is placed near the diaphragm. These form a parallel plate capacitor which is connected as one arm of a bridge, as shown in fig 2.12b. Any change in pressure causes a change in distance between the plates, which unbalance the bridge. The voltage output of the bridge corresponds to the pressure applied to the diaphragm.

Advantages: the following are the advantages of capacitive pressure transducers:

It gives rapid response to changes in pressure. Response times as short as 10 milliseconds are possible.

It can withstand a lot of vibrations and shock.

It is extremely sensitive.

It has good frequency response

Disadvantages: the following are the disadvantages of capacitive pressure transducers:

The metallic parts of the capacitor must be insulated from each other.

The performance of a capacitive transducer is adversely affected by dirt and other contaminants, because they change the dielectric constant.

Sensitivity of a capacitive transducer is adversely affected by changes in temperature.

Piezoelectric pressure transducers: these devices utilize the piezoelectric characteristics of certain crystalline and ceramic materials (such as quartz) to generate an electric signal. Such transducers utilize the principle that when pressure is applied on piezoelectric crystals, an electric charge is generated. There are about 40 crystalline materials that, when subjected to a squeeze generates an electric charge. Some of the piezoelectric materials are barium titanate sintered powder, crystal of quartz, tourmaline and Rochelle salts.

Construction and working: it consists of a diaphragm by which pressure is transmitted to the piezoelectric crystal Y_1 (fig.2.13). This crystal generates an electric charge which is amplified by a charge amplifier. A second piezoelectric crystal Y_2 is included to compensate acceleration of the device during use. This compensation is needed because rapid acceleration of the transducer creates additional pressure on the piezoelectric crystal.

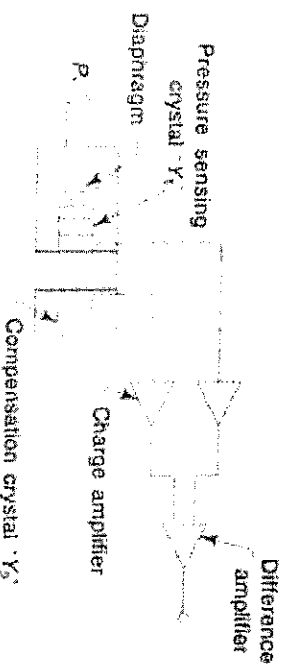


Fig 2.13 piezoelectric pressure transducer

Signals from the compensating crystal are amplified by a second charge amplifier. A different amplifier is used which subtract pressure alone; all effect of acceleration are removed.

Piezoelectric transducers are used to measure very high pressures that changes very rapidly. For example, the pressure inside the cylinder of a gasoline engine changes very rapidly from less than atmospheric pressure to many thousand pounds per square inch. Similar pressure changes occur in compressors, rocket motors etc. piezoelectric transducers may be used to measure pressures over ranges up to 0-50,000 psi. However, they cannot measure steady pressures. They respond only to changing pressure.

Advantages: the advantages of piezoelectric transducers are:

They need no external power and are therefore self-generating (active type).

It has very good frequency response.

Disadvantages: the advantages of piezoelectric transducers are:

They cannot measure static pressures.

The output of the transducer is affected by changes in temperature. Therefore, temperature compensating devices have to be used.

2.5 Calibration of pressure measuring instruments: pressure instrument calibration is a process of adjusting the instruments' output signal to match a known range of pressures. All instruments tend to drift from their last setting. This is because springs stretches, electronic components undergo slight changes on the atomic level, and other working parts sag, bend, or lose their elasticity.

Basic calibration procedure includes zero, span and linearity adjustments. Proper calibration provides the desired beginning and ending pressures, and produces an output signal

that is proportional to the process pressure. Calibration of the instrument is carried out by applying to it an air or liquid pressure whose value is accurately known.

2.6 Maintenance of pressure measuring instruments: Generally, instructions for the periodic maintenance of an instrument are given in the manuals supplied by the manufacturers, which explains when and how to service the instrument. Individual plants and industries also prepare their guidelines for periodic maintenance.

Periodic maintenance of pressure instrument generally includes the following three things:

Visual inspection: any leaks or damage in piping and wiring of the instrument can be examined visually and small corrections may be done during that period. But correction can be done without interrupting the operation of the plant and without causing safety hazards.

Blowdown and venting: Blowdown remove dirt and foreign materials from instrument line before they plug the line. By venting gas trapped in the liquid filled line, gas build up can be avoided, which also causes faulty pressure reading.

Cleaning and lubrication: instruments with mechanical linkages undergo wear and misalignment. Dirt may clog the linkages, causing the mechanism to stick. Checking the Linkages regularly reveals problems before they are serious enough to interfere with instruments operation.

The mechanical linkage should operate smoothly. If repair or adjustments are not needed, simple cleaning and lubrication of the instrument must be done according to manufacturer's specifications.

3.0 flow measurement

Flow measurements are made for fluids and sometimes for solids. It is made for determining the proportion and the amount of materials flowing in and out of a process. Without flow measurement, plant material balancing, quality control and even the operation of continuous process would be almost impossible.

3.1 Basic definitions:

- a) Mass flow rate ($\dot{m}$): this is the rate at which the mass of fluid passes a given cross-section in a unit time. $\dot{m} = \rho \Delta V$ (kg/s)
- b) Volume flow rate (V or Q): this is sometimes called the discharge and it refers to the rate at which the volume of fluid passes a given cross-section in a unit time. $Q = \Delta V$ (m^3/s)
- c) Viscosity: the frictional force within a liquid is known as viscosity of the liquid. The greater the viscosity, the less easy for it is for the fluid to flow and the stickier it feels. $\tau = \mu du/dy$.

The coefficient of dynamic friction μ can be defined as the shear force per unit area (shear stress) required to drag one layer of fluid with unit velocity pass another layer a unit distance from it in the fluid.

$$\mu = \frac{\tau du}{dy} \text{ (Ns/m}^2\text{)}$$

Kinematic viscosity V is defined as the ratio of the dynamic viscosity to mass density. Given by

$$V = \mu/\rho.$$

- d) Compressible and incompressible fluids: all fluids are compressible, so that their density will change with pressure, but under steady flow conditions and provided that the change in density is small, it is often possible to simplify the analysis of the problem by assuming that the fluid is incompressible and of constant density. That is to say the volume of the fluid cannot be reduced. Since liquids are relatively difficult to compress, it is usual to treat them as if they are incompressible in all cases of steady flow. Gases are easily compressed, that means that their volume is reduced under the effect of applied force.

3.2 Types of flow

- a) Laminar flow: a flow in which the particles of the fluid move in an orderly manner and retain the same relative cross-sections, i.e. the fluid particles moves in layers, each layer sliding over the adjacent one and the shear stress that arises is due to the action of the fluid viscosity only. All the streamlines are straight and therefore, the laminar flow is also called the viscous or the streamline flow. For example, the flow through a small diameter tube at low velocity, such as oil in tubes or conduits. Blood veins e.t.c.
- b) Turbulent flow: a flow in which the particles of the fluid move in a disorderly manner occupying different relative positions in successive cross-sections. Turbulent flow is also characterized by random and erratic movement of eddies or wakes. This eddy formation is responsible for high energy loss as compared to loss in laminar flow.

3.3.1 Reynolds number: this is a dimensionless number which is significant in the design of models of any system in which the effect of viscosity is important in controlling the velocities or the flow pattern of a fluid. It is the ratio of the inertia force to the viscous force

$$\begin{aligned} \text{Re} &= \text{inertia force/viscous force} \\ &= \rho v d / \mu \end{aligned}$$

Where, ρ = density of flowing fluid

v = average velocity

d = diameter of pipe

μ = absolute or dynamic viscosity

By performing an experiment with pipes of different sizes with water, Reynolds found that there is a transition from laminar to turbulent flow.

$Re < 2000$ = laminar flow

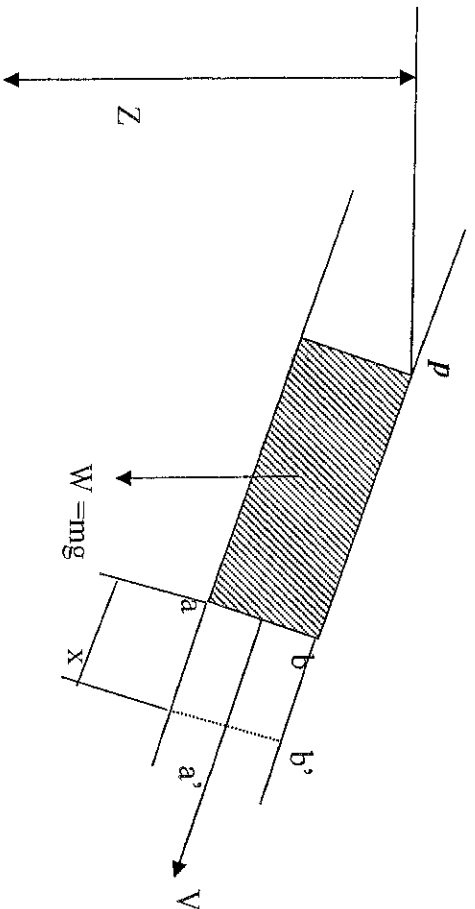
$Re > 4000$ turbulent flow

Re in between the two values is transition state.

3.2.2 Continuity of flow: continuity of flow embodies the principle of conservation of mass according to which fluid matter is neither created nor destroyed. For steady flow the mass of fluid in the control volume remains constant and the relationship reduces to:

Mass of fluid entering per unit time = mass of fluid leaving per unit time.

3.3 Forms of energy possessed by a moving fluid:



Energy of a flowing fluid such as shown above will possess potential energy (P.E.) due to its height Z above datum level and kinetic energy (K.E.) due to its velocity V .

For an element of weight $W = mg$

$$P.E. = mgZ$$

$$PE \text{ per unit weight} = mgZ/mg = Z$$

$$\text{Kinetic energy K.E.} = \frac{1}{2} mv^2$$

$$KE \text{ per unit weight} = \frac{1}{2} mv^2/mg = \frac{v^2}{2g}$$

A steadily flowing fluid can also do work because of its pressure. At any given cross-section, the pressure generates a force and as the fluid flows the cross-section will move forward and so work will be done. If at the section is P and the area of cross-section is A

$$\text{Force exerted on } ab, F = PA$$

After a weight mg of fluid has flowed along the stream tube section, ab would have moved to a new position $a'b'$

$$\text{Volume of } ab = m/\rho \quad (w = \rho gV)$$

$$\therefore \text{Distance } aa' (x) = m/\rho A \quad \text{dist} = \text{vol/area}$$

$$\text{Work done} = \text{force} \times \text{distance } aa'$$

$$= P A \times m/\rho A = P m/\rho$$

Work done per unit weight = $P m/\rho m g = P/\rho g$ (pressure energy of a fluid flowing under pressure)

Total energy per unit weight = pressure energy per unit weight + KE per unit weight + PE per unit weight

$$H = P/\rho g + V^2/2g + Z \dots\dots\dots \text{Bernoulli's equation}$$

Between any two points, subscript 1 and 2 are added and the equation becomes

$$P_1/\rho g + v_1^2/2g + Z_1 = P_2/\rho g + v_2^2/2g + Z_2$$

Bernoulli's theorem states that: the total energy of each particle of a body of fluid is the same provided that no energy enters or leaves the system at any point. The division of this energy between potential, pressure and kinetic energy may vary, but the total still remain constant. Or

For a steady flow of frictionless fluid along a streamline, the total energy per unit weight remains constant from point to point although its division between the three forms of energy may vary.

3.4 Methods of flow measurement:

Many accurate and reliable methods are available for measuring flow, some of which are applicable only to liquids, some only to vapour and gases and some others to both. Fluid measured may be clear or opaque, clean or dirty, wet or dry, erosive or corrosive. Fluid stream may be multiphase, liquid, vapour or slurry. The flow may be laminar or turbulent, Viscosities may vary, pressure may change from vacuum to many atmospheres, and temperature may range from a few degrees to hundreds of degrees. Flow rates may range from few drops per hour to thousands of gallons per minute. Thus with this possible assessment of conditions and requirements, the various flow measuring methods can be divided into three groups:

Inferential type flow meters

Quantity flow meters

Mass flow meters

3.4.1 Inferential flow measurement: in the inferential type flow measuring methods, the flow rate is inferred from a characteristics effect of a related phenomenon. A classic example of an inferential measurement is the human body running fever, resulting from a serious sickness, in which the rise in body temperature (fever) is not the cause of sickness but a characteristics phenomenon resulting from the ailment. The following are the inferential type flow measuring methods:

Variable head or differential meters

Variable area meters

Magnetic meters

Turbine meters

Target meters

Thermal meters

Vortex meters

Ultrasonic flow meters

Variable head or differential flow meters: this is one of the oldest and most widely used methods of industrial flow measurement. It measures volume rather than mass flow rate. These flow meters operate on the principle that a restriction (or obstruction) in the line or pipe of a flowing fluid, introduced by an orifice plate or a venturi tube or elbow produces a differential pressure across the restriction element which is proportional to the flow rate. The relationship is not linear but it has a square root relationship in that the flow rate is proportional to the square root of the differential pressure.

Advantages of differential flow meters:

Its cost is relatively low especially for large lines.

It offers the widest applicational coverage of any type of matter.

It is accurate and reliable.

It can easily be removed without shutting down the process.

It is adaptable to any pipe size and flow rate.

Disadvantages and limitations:

There is relatively high permanent pressure loss in it.

It is difficult to use for slurry service.

It is difficult to measure pulsating flow with this flow meter.

Its accuracy is dependent on many fixed fluid characteristics such as temperature, pressure, compressibility etc.

Part of a differential flow meter: a differential flow meter is basically consists of two parts:

Primary elements

Secondary elements

The parts of the meter used to restrict the flow in the pipeline in order to produce a differential pressure are known as primary elements. They are listed below:

Orifice plates

Venturi tubes

Flow nozzles

Dall tubes

Pitot tubes

Annubar tubes

Elbow taps

Weir

Flume

Secondary elements are those which measure the differential pressures produced by the primary elements and convert them to usable force or signals. Various secondary elements are listed below:

Manometers

Bellow meters

Force balance meters

Ring balance meters.

These secondary elements are the same as those discussed in pressure measurement.

Orifice plates: these are the cheapest and simplest form of primary elements and are used more frequently than the other types. An orifice plate is inserted in the line and the differential pressure across it is measured. There are four types of orifice plates which are listed below:

Concentric orifice plate

Eccentric orifice plate

Segmental orifice plate

Quadrant edge orifice plate

The figure below shows the various types of orifice plates.

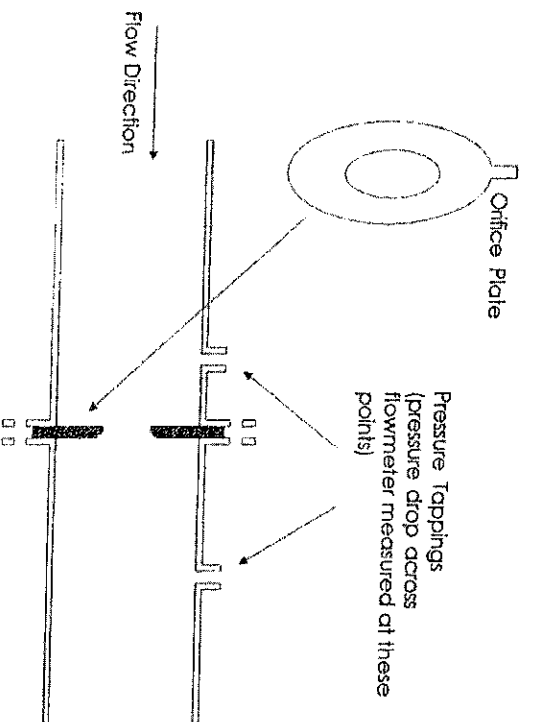


Fig. 3.2: orifice plate



Fig. 3.3: different configuration of orifice plates

Concentric orifice plate: it is the most widely used. It is usually made of stainless steel and its thickness varies from 3.175 to 12.70 mm depending on pipe size and flow velocity. It has a circular hole (orifice) in the middle, and is installed in the pipeline with the hole concentric to the pipe. It is also made from other materials such as nickel, monel, phosphor bronze etc, to withstand the corrosive effect of the fluid. The plate thickness at the orifice edge should not exceed any of the following

$$D/50; d/8; (D-d)/8$$

Where D = the pipe internal diameter, d = the orifice diameter or bore

Eccentric orifice plate: it is similar to the concentric plate except for the offset hole which is bored tangential to a circle, concentric with the pipe and of diameter equal to 98% of that of the pipe. Location of the bore prevents damming of solid materials or foreign particles and makes it

useful for measuring fluids containing solids, oils containing water and wet steam. The concentric orifice plate is used where liquid fluid containing a relatively high percentage of dissolved gases, and is installed with the bore tangential to the upper surface of the pipe when the flowing material is liquid and tangential to the lower surface of the pipe when the fluid is gas.

Segmented orifice plate: this type of plate is used for the same type of service as the eccentric orifice plate. It has a hole which is a segment of a circle, the diameter of which is customarily 98% of the pipe diameter. It is installed with the dam horizontal and with the curved section of the opening coincidental with the lower surface of the pipe.

Quadrant edge orifice plate: this type of orifice plate is used for flow such as heavy crudes, syrups, and slurries, and viscous flow. It is constructed in such a way that the edge is founded to form a quarter-circle. The plate has a concentric opening with a rounded-up stream edge rather than the sharp, square edge normally used. It may be used when the line Reynolds number range from 100,000 or above, down to 3,000 to 5,000 with coefficient accuracy of approximately 0.5%.

Advantages of orifice plates:

Low cost.

Can be used in a wide range of pipe size (3.175 to 1828.8 mm).

Can be used with different pressure devices.

Well known and predictable characteristics.

Available in many materials.

Disadvantages of orifice plates:

Cause relatively high-pressure loss.

Tend to clog, thus reducing use in slurry services.

Have square root characteristics.

Accuracy dependent on care during installation.

Venturi tubes: a venturi tube is used when permanent pressure lost is of great importance, and where maximum accuracy is desired in the measurement of high viscous fluids. It consists of a straight inlet section of the same diameter as the pipe and in which the high pressure tap is located, and a converging conical inlet section in which the cross section of the stream decreases and the velocity increases with a consequent increase of velocity head and decrease of pressure head, a cylindrical throat which provide for the low pressure tap location of the decreased pressure in an area where flow velocity is neither increasing nor decreasing, and a diverging recovery cone in which the velocity decreases and the decreased velocity head is recovered as pressure head, as shown in fig 3.3 below. The pressure taps are located one quarter to one half pipe diameter upstream of the inlet cone and at the middle of the throat section.

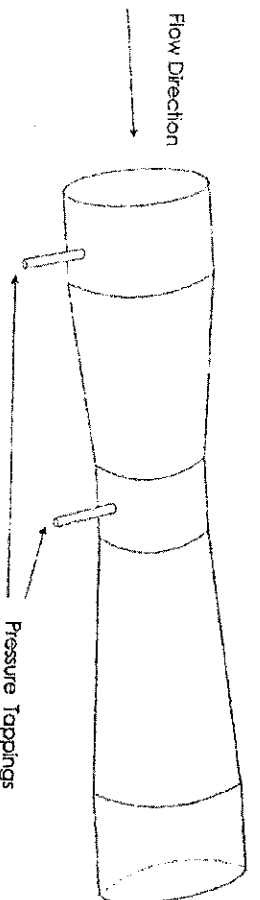


Fig. 3.4: venturi tube

The venturi tube can be used to handle fluid that can be handled by an orifice plate and fluids that contains some solids, because these venturi tubes contain no sharp corners and do not project into the fluid stream. It can also be used to handle slurries and dirty liquids that build up around other primary elements, if the pressure taps are protected from plugging.

Venturi tubes are made of cast iron or steel, and are built in many forms. They are available in sizes from 100 mm to 813 mm.

Advantages:

Cause low permanent pressure lost.

Widely used for high pressure rates

Available in very large sizes.

Has well known characteristics.

More accurate over wide flow ranges than orifice plate or nozzles.

Disadvantages:

High cost.

Generally, not useful below 76.2 mm pipe size.

More difficult to inspect due to its construction.

Flow nozzles: the flow nozzles are used for flow measurement at high flow velocities and are more rugged and more resistant to erosion than the sharp edged orifice plate. It consists of a convergent inlet whose shape is quarter ellipse, and a cylindrical throat, as shown in fig 3. Differential pressure measuring taps are usually located one pipe diameter upstream and one and a half diameter downstream from the inlet face of the nozzle.

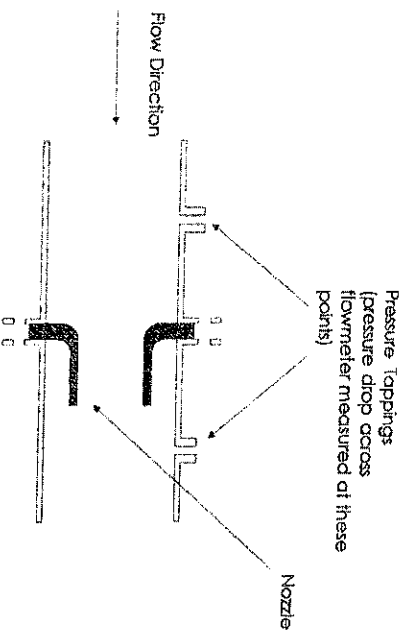


Fig. 3.5: flow nozzle

Flow nozzles are manufactured commonly from materials such as stainless steel or chrome-moly steel and should be used at Reynolds number of 50,000 or above. However, data is available for Reynolds number down to 6,000; so, it is possible to use flow nozzles to with more viscous fluids.

Advantages:

Permanent pressure loss lower than that for an orifice plate.

Available in numerous materials.

For fluids containing solid that settle.

Widely accepted for high pressure and temperature steam flow.

Disadvantages:

Cost is higher than orifice plates.

Limited to moderate pipe size.

Requires more maintenance.

Weirs: weirs are used to measure flow rate primarily in open channels such as water works including irrigation, waste and sewage system, and in pipes and conduits that are generally not completely filled with liquid. It is an obstruction in a flowing stream over which the liquid is made to pass. With the use of weir, flow rates can be measured from a few gallons per minute to millions of gallons per day. There are three types of weirs: rectangular, V-notch and trapezoidal, as shown in fig 3.5

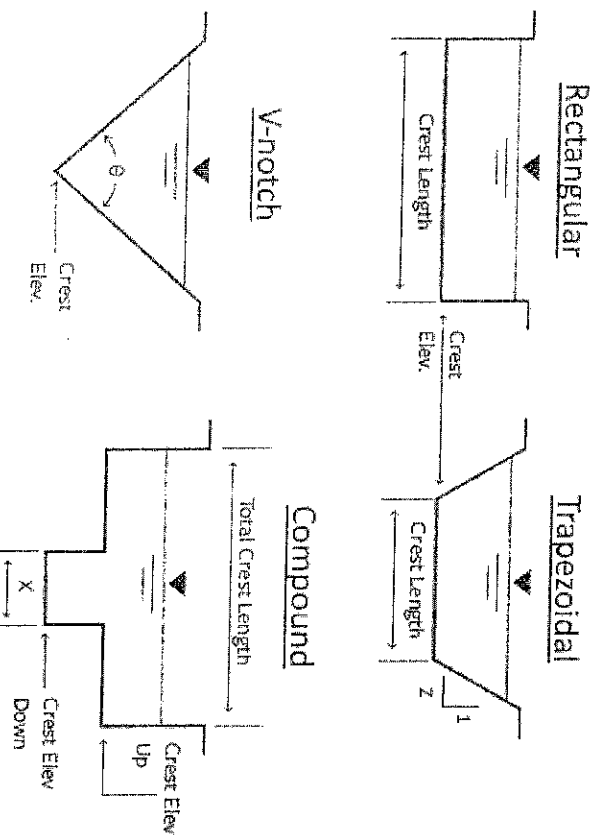


Fig. 3.6: different types of weirs

Advantages:

Low cost.

Can be constructed on location being used.

Not easily damaged

Disadvantages:

Applicable only to open channels measurement.

Field calibration required.

Poor accuracy.

Quantity flow meters:

These are used for the measurement of small percentage of industrial flow rate. These meters operate by passing the fluid to be measured through the meter in separate and distinct increments of alternately filling and emptying containers of known fixed capacity. The number of times the container is filled and emptied gives the quantity of flow.

ROTAMETER

The rotameter is the most extensively used form of the variable area flowmeter. It consists of a vertical tapered tube with a float which is free to move up or down within the tube as shown in Fig.

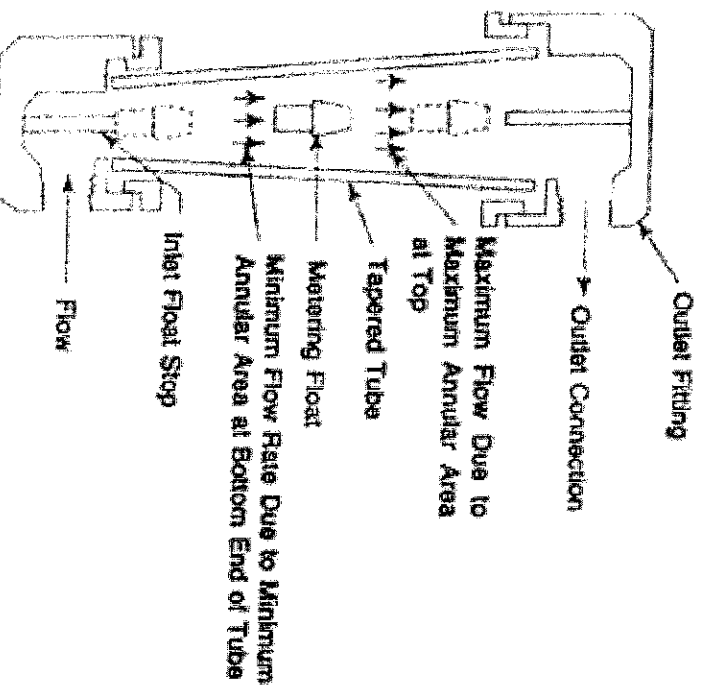


Fig. 3.7: A Rotameter

The tube is made tapered so that there is a linear relationship between the flow rate and position in the float within the tube

The free area between float and inside wall of the tube forms an annular orifice. The tube is mounted vertically with the small end at the bottom. The fluid to be measured enters the tube from the boom and passes upward around the float and exit at the top.

When there is no flow through the rotameter, the float rests at the bottom of the metering tube where the maximum diameter of the float is approximately the same as the bore of the tube. When fluid enters the metering tube, the float moves up, and the flow area of the annular orifice increases.

The pressure differential across the annular orifice is parabolical to the square of its flow area and to the square of the flowrate. The float is pushed upward until the 'limns force produced by the pressure differential across it super and lower surface is equal to the weight of the float. If the flowrate rises, the pressure differential and hence the lining force increases temporarily, and the float then rises, widening the annular orifice until the force cawed by the pressure differential is again equal to the weight of the Boat.

Thus, the pressure differential remains constant and the area of the annular orifice (i.e., free area between float and inside wall of the tube) to which the float moves. Changes in proportion to the flowrate. Any decrease in flowrate causes the float to drop to a lower position. Every float position corresponds to one particular flow rate for a quid of a given density and viscosity.

Advantages of Rotameter

- Simplicity of operation.
- Ease of reading and installation.
- Relatively low cost.

• Handles wide variety of corrosive fluids.

- Easily equipped with data transmission, indicating and recording devices.

Disadvantages of Rotameter: -

- Glass tube subject to breakage.
- Limited to small pipe sizes and capacities.
- Less accurate compared to venture and orifice meters.
- Must be mounted vertically.
- Subject to oscillations

TURBINE FLOW METER

Principle: The permanent magnet attached to the body of rotor is polarized at 90° to the axis of rotation. When the rotor rotates due to the velocity of the fluid (V), the permanent magnet also rotates along with the rotor.

Therefore, a rotating magnetic field will be generated which is then cut by the pickup coil. Due to this ac-voltage pulses are generated whose frequency is directly proportional to the flowrate.

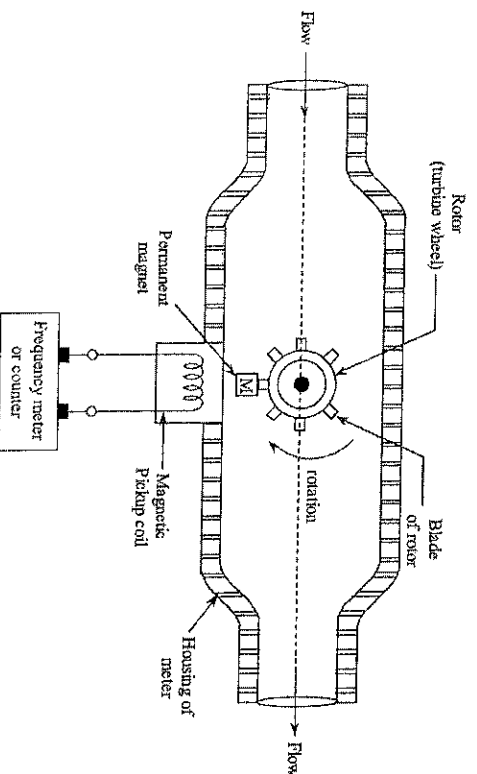


Fig. 3.8: A Turbine flow meter

Advantages:

- Good accuracy and repeatability. Easy to install and maintain.
- Low pressure drop.
- Electrical output is available.
- Good transient response.

Disadvantages:

- High cost.
- The bearing of the rotor may subject to corrosion.
- Wear and tear problems.

Applications:

- Used to determine the fluid flow in pipes and tubes.
- Flow of water in rivers.
- Used to determine wind velocity in weather situations or conditions.

Thermal flow meter:

When a fluid flows over an electrically heated surface, heat transfer takes place from the surface or wire to the fluid. Hence, the temperature of the heated wire decreases which causes variations in the resistance. The change that occurred in the resistance of the wire is related to the flowrate. The sensor is a 5-micron diameter platinum tungsten wire welded between the two prongs of the probe and heated electrically as a part of Wheatstone bridge circuit.

When the probe is introduced into the fluid flowing, it tends to be cooled by the instantaneous velocity and consequently there is a tendency for the electrical resistance to change. The rate of cooling depends upon the dimensions and physical properties of the wire. Difference of the temperature between the wire and fluid, physical properties of the fluid, string velocity undermeasurement.

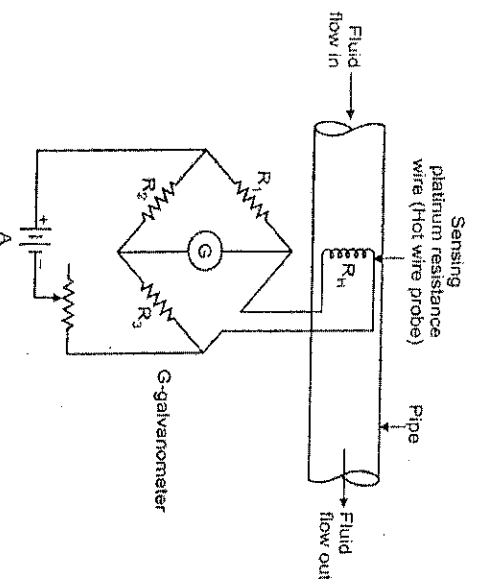


Fig. 3.9: thermal flow meter in a bridge setup

MAGNETIC FLOWMETER

Magnetic flow meter depends upon the faradays law of electromagnetic induction.

These meters utilize the principles of faradays law of electromagnetic induction for making a flow measurement.

It states that whenever a conductor moves through a magnet field of given field strength; a voltage is induced in the conductor, which is proportional to the relative between the conductor and the magnetic field.

CONSTRUCTION & WORKING

In case of magnetic flow meters electrically conductive flowing liquid works as the conductor the induced voltage.

The length L of which is the distance between the electrodes and equals the pipe diameter.

As the liquid passes through the pipe section, it also passes through the magnetic field setup by the magnet coils, thus inducing the voltage in the liquid which is detected by the pair of electrodes mounted in the pipe wall.

The amplitude of the induced voltage is proportional to the velocity of the flowing liquid.

The magnetic coils may be energized either by AC or DC voltage, but the recent development is the pulsed DC-type in which the magnetic coils are periodically energized.

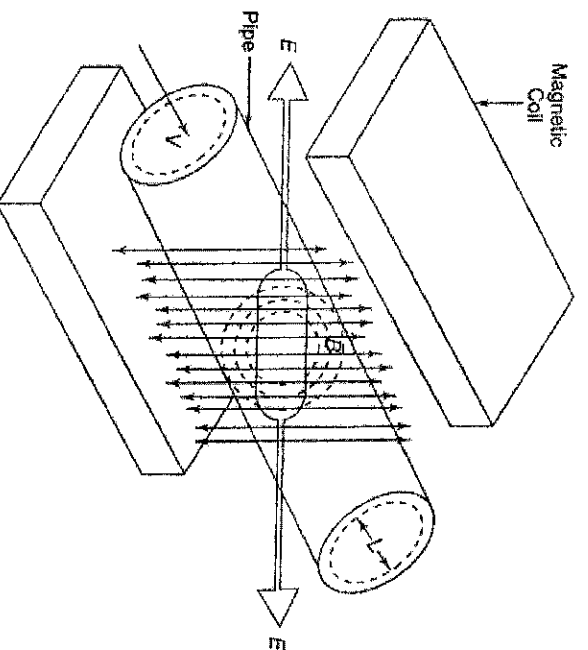


Fig. 3.10: A Magnetic flow meter

Advantages:

- It can handle greasy materials.
- It can handle corrosive fluids.
- Accuracy is good.
- It has very low pressure drop.

Disadvantages:

- Cost is more.
- Heavy and larger in sizes.
- Explosion proof when installed in hazardous electrical areas.
- It must be full at all times.

ULTRASONIC FLOW METER

The velocity of propagation of ultrasonic sound waves in a fluid is changed when the velocity of the flow of fluid changes. The arrangement of flow rate measurement using ultrasonic transducer contains two piezo-electric crystals placed in the fluid whose flow rate is to be measured of these two crystals one acts as a transmitting transducer and the other acts as a receiving transducer.

The transmitter and receiver are separated by some distance say " L ". Generally, the transmitting transducer is placed in the upstream and it transmits ultrasonic pulses.

These ultrasonic pulses are then received by the receiving transducer placed at the downstream flow. Let the time taken by the ultrasonic pulse to travel from the transmitter and received at the receiver is " Δ ".

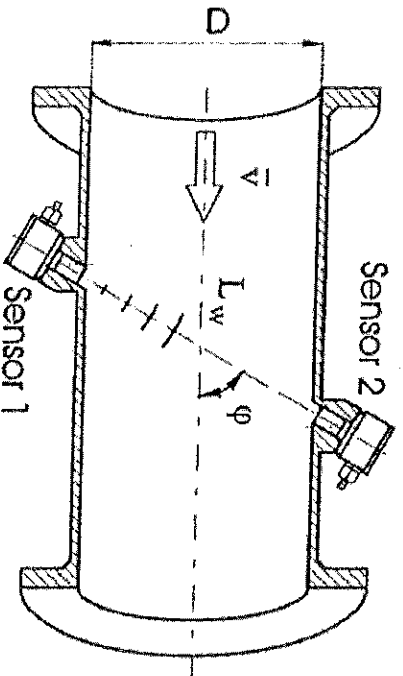


Fig. 3.11: Ultrasonic flow meter

Mass flow meter: the mass flow meter utilizes the principle of angular momentum. These are rarely found in industrial application today. Commercially available mass flow devices include impeller-turbine and twin turbine types and are operating on the principle of gyroscope. Industrial use of these types of meters are limited because of high cost and relative ease and convenience of calculation by other methods.

4.0 liquid level measurements

Level measurement is one of the oldest of all measurements. The measurement of liquid level is of great importance in the industrial field. The level of liquid may affect both pressure and the rate of flow in and out of the tank or vessel; hence, the quantity may be affected.

4.1 Methods of liquid level measurement: generally, there are two methods used in industries for measuring liquid level. They are:

Direct methods

Indirect methods

4.1.1 Direct methods: this is the simplest method of measuring liquid level. In this method, the liquid level is measured directly by means of the following level indicators:

Hook-type level indicator

Sight glass

Float-type

Sight glass: a sight glass (also called a gauge glass) is used for continuous indication of liquid level within the tank or vessel.

Construction and working: a sight glass instrument consists of a graduated tube of toughened glass which is connected to the interior of the tank at the bottom in which the water level is required. Fig 4.1 shows a simple sight glass for an open tank in which the liquid level in the sight glass matches the level of liquid in the tank. As the level of the liquid in the tank rises and falls, the level in the sight glass also rises and falls accordingly. Thus, by measuring the level in the sight glass, the level of liquid in the tank is measured.

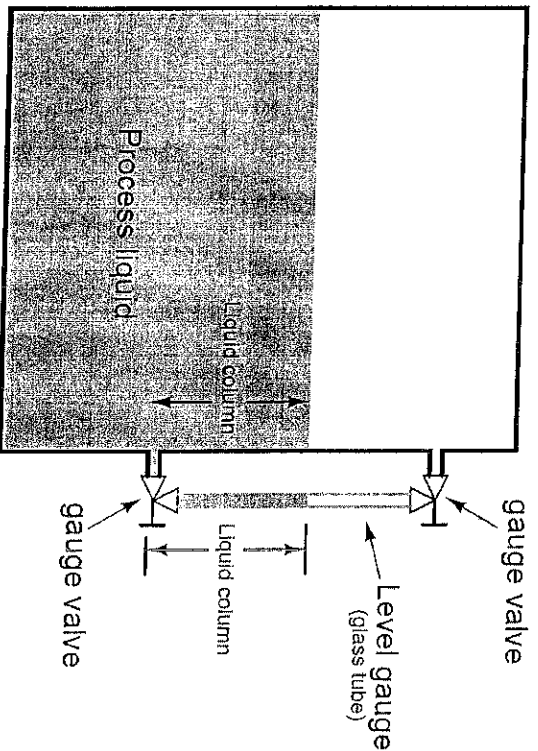


Fig. 4.1: Sight glass

When it is desired to measure the liquid level with the liquid under pressure or vacuum, the sight glass must be connected to the tank at the top as well as at the bottom, otherwise the pressure difference at the tank and at the sight glass will cause a false reading. In this case, the glass tube is enclosed in a protective housing, and two valves are provided for isolating the gauge from the tank in case of breakage of the sight glass. The standard practice is not to go for a glass tube of more than 900 mm length. In case the height of the tank is more than 900 mm, two or more sight glass levels are provided at different levels. This type of gauge is made to withstand pressures of 350 psi of steam at 252°C, or 1000 psi of liquid pressures.

Advantages:

Direct reading is possible.

Glasses designs are available in numerous materials for corrosion resistance.

Disadvantages:

- Readings are taken at the site where the tank is located.
- The liquid in the sight glass may freeze in cold weather even if the liquid inside the tank does not.
- Overlapping gauges needed for long level spans.
- Accuracy and readability depend on the cleanliness of the glass and the fluid.

Float type level indicator: this type of level indicator is used to measure liquid levels in tank in which, a float rests on the surface of liquid and follows the changing level of liquid. The movement of the float is transmitted to the pointer through a suitable mechanism which indicates the level on a calibrated scale. Various types of floats are used such as hollow metal spheres, cylindrical-shaped floats and disc-shaped floats.

Construction and working: a simple form of float operated mechanism for continuous liquid level measurement is shown in fig 4.2. In this case, the movement of the float is transmitted to the pointer by a stainless steel or phosphor-bronze flexible cable wound round a pulley, and the pointer indicates the level of liquid in the tank. The float is made of corrosion resisting material (such as stainless steel) and rest on the liquid level surface between two grids to avoid error due to turbulence.

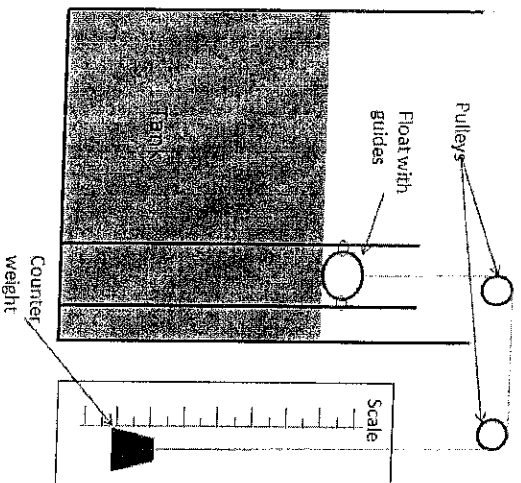


Fig. 4.2: float type level indicator

With float operated type of mechanism, the liquid level can be transmitted to a distant place using a hydraulic transmission system. It consists of four bellow elements, two in the transmitter side (C and D) and two in the receiver side (A and B) and are fixed at the outer ends. Bellows B, C and A, D are hydraulically connected through pipes filled with oil. When the float moves up or down according to change in level, its position is transmitted by a lower arm to the bellows assembly A – B. when the level rises, B is compressed and A expands, thus causing oil in the pipe to flow from B to C, and from D to A. any change in temperature affects both bellows, i.e. both expand and contracts. Therefore, level transmission at the pointer is not affected. With this type of instruments, level transmission up to 6.35m can be achieved.

Advantages:

It is possible to read the liquid level in a tank from the ground level even if the tank is kept below the ground level.

Its cost is low and has a reliable design.

It operates over a large temperature range.

Available in a wide range of corrosion resistant materials.

Disadvantages:

They are normally limited to moderate pressures.

They are tailored to tank geometry,

For the measurement of viscous liquid, a liquid seal is incorporated as shown in fig 4.4

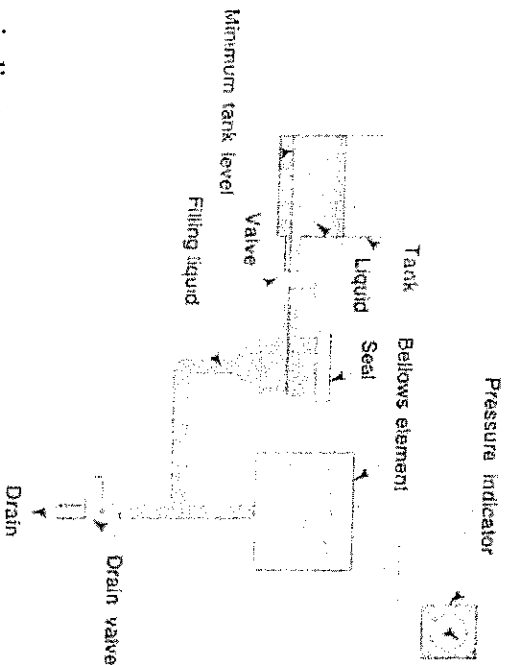


Fig. 4.3: air bellow indicator

Electrical methods: there are many electrical methods for measuring and indicating liquid level in industries. Of these the following are the most commonly used in industries:

Capacitance level indicators

Radiation level indicators

Capacitance level indicator: the principle of capacitance level indicator is based upon the familiar capacitance equation of a parallel plate capacitor given by:

$C = K \frac{A}{D}$, where, C = capacitance, in farad; K = dielectric constant; A = area of plate, in m^2 ; D = distance between two plates, in m.

Therefore, it is seen from the above equation that if A and D are constant, then the capacitance of a capacitor is directly proportional to the dielectric constant, and this principle is utilized in the capacitance level indicator.

Construction and working: fig 4.4 show a capacitance type liquid level indicator. It consists of an insulated capacitance probe (which is a metal electrode) firmly fixed near and parallel to the metal wall of the tank. If the liquid in the tank is non inductive, the capacitance probe and the tank wall form the plates of a parallel plate capacitor and the liquid in between them acts as the dielectric. If the liquid is conductive, the capacitance probe and the liquid form the plate of the capacitor while the insulation of the probe acts as dielectric. A capacitance measuring device is connected with the probe and the wall of the tank, which is calibrated in terms of the level of liquid in the tank.

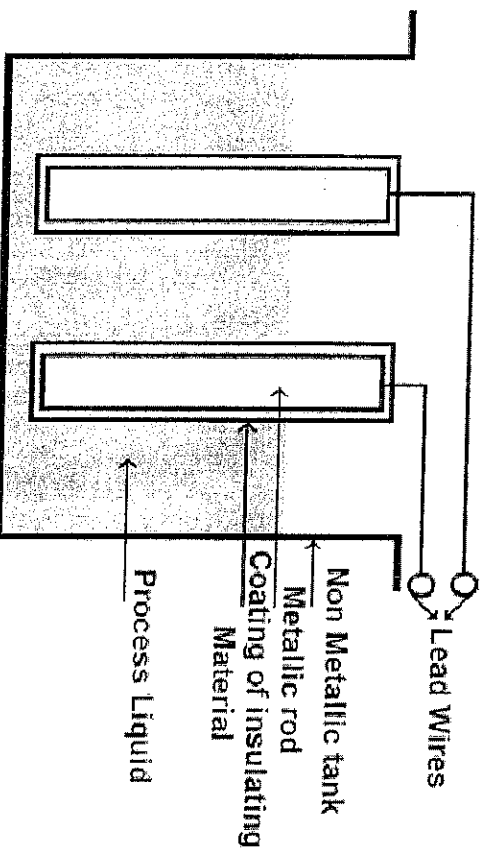


Fig. 4.4: Capacitive level indicator

When the level of the liquid in the tank rises, the capacitance increases. When the liquid level decreases, the capacitance also decreases. This increase and decrease in the capacitance is measured and is displayed on the indicator calibrated in terms of liquid level

4.1.2 Indirect methods: the following are the indirect methods of liquid level measurement generally used in industries:

Hydrostatic pressure type

Electrical methods

Hydrostatic pressure type: a liquid in a tank at rest exerts a force on the walls of the tank. This force in a liquid at rest, is known as hydrostatic pressure, and is proportional to the depth of liquid in the tank. Hydrostatic pressure methods used for liquid level measurement are listed below:

Pressure gauge method

Air bellows

Air purge system

Liquid purge system

Pressure gauge method: this is simplest method used for liquid level measurement in an open tank. Hydrostatic pressure of any liquid in an open tank is given by the equation:

$P = \rho \times h \times S_g$ or $h = P/\rho \times S_g$, where, P = pressure in psi or N/m^2 , ρ = density of water; S_g = specific gravity; h = height of liquid (often called head) in inch or meter.

Hydrostatic pressure of a liquid in a closed tank or vessel is given by the equation:

$$P = (\rho \times h \times S_g) + \text{external pressure on liquid.}$$

Therefore, the pressure measured at the bottom of a tank containing a liquid of known density and specific gravity is directly proportional to the height or level of liquid in the tank.

Construction and working: this consist of a pressure gauge connected at the lowest level of the tank. The level at which the pressure gauge is fitted is known as the reference level and the static pressure measured by the gauge is the measure of the height of the liquid column above the reference level and hence the liquid level. A liquid seal is connected the piping on the tank including shut off valve while measuring corrosive or highly viscous liquids. This liquid seal consists of a fluid with which the measuring system is filled. This filling liquid transmits the pressure head of the measured liquid. The free surface of the filling liquid is kept in indirect contact with the measured liquid. These two liquids should not be allowed to mix or react chemically.

The location of the pressure gauge should be chosen carefully, since any difference in elevation above or below the lowest level of the tank will affect the reading.

Air bellows: air bellows are used for liquid level measurement where an indicator cannot be conveniently located at the specified datum line.

Construction and working: air bellows consist of the bellow element which is connected by the tubing with the pressure indicator. Air is sealed in the cavity above the bellows and inside the tubing to the pressure indicator. When the tank is empty, the sealed air is uncompressed and corresponds to zero on the pressure indicator. As the tank is filled with liquid, the head of the liquid in the tank flexes the bellows, which compresses the air in the bellows. The compression of sealed air is transmitted to the indicator which is calibrated in the tank liquid level.

Advantages:

It is very useful in a small system.

It is very sensitive.

There are no moving parts exposed to the fluid.

It is suitable for continuous indication/ or control

Remote adjustment of span and zero is possible in this type of level indicator.

It is good for use with slurries.

Probe materials for most corrosive fluid are available.

Disadvantages:

The performance of a capacitive level indicator is severely affected by dirt and other contaminants, because they change the dielectric constant.

Its sensitivity is adversely affected by change in temperature.

Measured fluid must have proper dielectric qualities.

They usually require recalibration if measured material changes.

Probe length and mounting must suit the tank.

Radiation level detector: these are used where other electrical methods would not survive. Another reason is it does not need to come in contact with the liquid being measured.

Construction and working: This consist of a gamma rat holder on one side of the tank and a gamma detector on the other side of the tank (fig 4.5). The gamma rays from the source are directed towards the detector in a thin band of radiation. When the gamma rays penetrate the thick walls of the tank, its energy level afterwards is greatly reduced. The radiation received at the gamma detector is inversely proportional to the thickness of the tank walls and the medium between the radiation source and the detector. That is the thicker the medium between source and detector, the less the radiation received by the detector and vice versa.

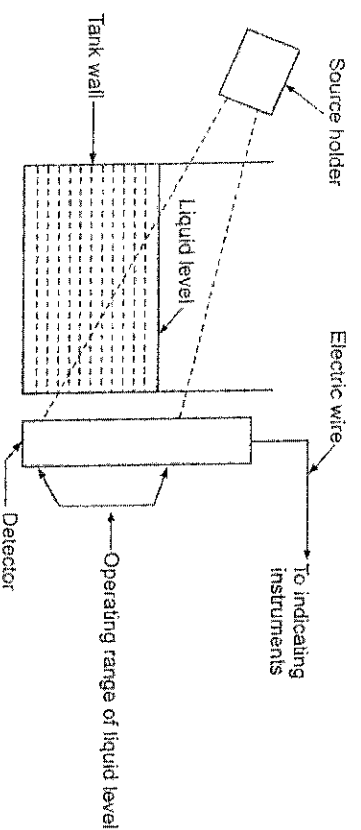


Fig. 4.5: radiation level detector

When the tank is empty, the gamma ray passes only through the two tank walls and the air or vapour in the empty tank. When liquid enters the tank and its level rises, the radiation beam passes through a path in the liquid as well as the tank walls. The liquid in the tank reduces the radiation received by the detector. The amount of radiation received is inversely proportional to the amount of liquid between the radiation source and the detector. The difference in the amount of radiation received by the detector corresponds to the liquid level. Thus, when the liquid level rises, the amount of radiation received is reduced and vice versa. The radiation loss received by the tank walls is constant whether the tank is empty or full.

Advantages:

There is no physical contact with the liquid.

They are suitable for molten metal as well as liquids of all types

They are useful at very high temperature and pressure.

They have good accuracy and response.

They have no moving parts.

Disadvantages:

Reading is affected by change in liquid density.

Radiation source holder may be heavy.

Their cost is relatively high.